PRACTICE SHEET

1. What is the product of first 2n + 1 terms of a geometric progression?
(a)The (n + 1)th power of the nth term of the GP
(b)The (2n + 1)th power of the nth term of the GP
(c)The (2n+1) the power of the (n+1)th term of the GP
(d)The nth power of the (n + 1)th terms of the GP

2. If x + 1, 4x + 1, and 8x + 1 are in geometric progression, then what is the non – trivial value of x?
(a) – 1 (b) 1
(c) 1/8 (d) 1/4

3. The equation $(a^2 + b^2) x^2 - 2b (a + c) x + (b^2 + c^2) = 0$ has equal roots. Which one of the following is correct about a, b, and c?
(a) They are in AP
(b) They are in GP
(c) They are in HP
(d) They are neither in AP, nor in GP, nor in HP

4. For an AP with first term u and common difference v, the p^{th} term is 15 uv more than the q^{th} term. Which one of the following is correct?
(a) p = q + 15v (b) p = q + 15 u
(c) p = q + 14 v (d) p = q + 14 u

5. If a, b and c are three positive numbers in an arithmetic progression, then :
(a) $ac > b^2$ (b) $b^2 > a + c$
(c) $ab + bc \leq 2ac$ (d) $ab + bc \geq 2ac$

6. If 1, x, y, z, 16 are in geometric progression, then what is the value of x + y + z?
(a) 8 (b) 12
(c) 14 (d) 16

7. If the nth term of an arithmetic progression is 3n + 7, then what is the sum of its first 50 term?
(a) 3925 (b) 4100
(c) 4175 (d) 8200

8. What is the value of $9^{1/3}, 9^{1/19}, 9^{1/27}, \ldots\ldots\infty$?
(a) 9 (b) 3
(c) 91/3 (d) 1

9. After paying 30 out of 40 installments of a debt of Rs. 3600, one third of the debt is unpaid. If the installments are forming an arithmetic series, then what is the first installment?
(a) Rs. 50 (b) Rs. 51
(c) Rs. 105 (d) Rs. 110

10. The difference between the nth term and (n–1)th term of a sequence is independent of n. Then the sequence follows which one of the following?
(a) AP (b) GP
(c) HP (d) None of these

11. If the nth term of an arithmetic progression is $2n-1$, then what is the sum upto n terms?
(a) n^2 (b) $n^2 - 1$
(c) $n^2 + 1$ (d) $\frac{1}{2}n(n+1)$

12. Sum of first n natural numbers is given by $\frac{n(n+1)}{2}$ .What is the geometric mean of the series 1, 2, 4, 8, ... 2^n?
(a) 2^n (b) $2^{n/2}$
(c) $2^{1/2}$ (d) 2^{n-1}

13. If the number of terms of an A.P. is $(2n + 1)$, then what is the ratio of the sum of the odd terms to the sum of even terms?
(a) $\frac{n}{n+1}$ (b) $\frac{n^2}{n+1}$
(c) $\frac{n+1}{n}$ (d) $\frac{n+1}{2n}$

14. If he sum of 'n' terms of an arithmetic progression is $n^2 - 2n$, then what is the n^{th} term?
(a) $3n - n^2$ (b) $2n - 3$
(c) 2n + 3 (d) 2n – 5

15. If a, 2a + 2, 3a + 3 are in GP, then what is the fourth term of the GP?
(a) –13.5 (b) 13.5
(c) –27 (d) 27

16. What is sum to the 100 terms f the series 9 + 99 + 999 + ...?
(a) $\frac{10}{9}(10^{100}-1)-100$
(b) $\frac{10}{9}(10^{99}-1)-100$
(c) $100(100^{10}-1)$
(d) $\frac{9}{100}(10^{100}-1)$

17. Natural numbers are divided into groups as (1), (2,3), (4,5,6), (7, 8, 9, 10) and so on. What is the sum of the numbers in the 11^{th} group?
(a) 605 (b) 615
(c) 671 (d) 693

18. If the mth term of an A.P. is (1/n) and the nth term is (1/m), then its $(mn)^{th}$ term is:
(a) –mn (b) –1
(c) 1 (d) 1/mn

19. If x^2, y^2, z^2 are in A.P., then y + z, z+x, x+y are in:
(a) A.P. (b) H.P.
(c) G.P. (d) None of these

20. If the number of terms of an A.P. is (2n+1), then what is the ratio of the sum of the odd terms to the sum of even terms?
(a) $\frac{n}{n+1}$ (b) $\frac{n^2}{n+1}$
(c) $\frac{n+1}{n}$ (d) $\frac{n+1}{2n}$

21. A square is drawn by joining mid-points of the sides of a square. Another square is drawn inside the second square in the same way and the process is continued indefinitely. If,

the side of the first square is 16cm, then what is the sum of the areas of all the squares?
(a) 256sq. cm (b) 512sq. cm
(c) 1024sq. cm (d) 512/3sq. cm

22. If the A.M. and G.M. between two numbers are in the ratio m:n, then what is the ratio between the two numbers?
(a) $\frac{m+\sqrt{m^2-n^2}}{m-\sqrt{m^2-n^2}}$ (b) $\frac{m+n}{m-n}$
(c) $\frac{m^2-n^2}{m^2+n^2}$ (d) $\frac{m^2+n^2-mn}{m^2+n^2+mn}$

23. The sum of an infinite geometric progression is 6. If the sum of the first two terms is 9/2, then what is the first term?
(a) 1 (b) 5/2
(c) 3 or 3/5 (d) 9 or 3

24. The arithmetic mean of two numbers exceeds their geometric mean by 2 and the geometric mean exceeds their harmonic mean by 1.6. What are the two numbers?
(a) 16, 4 (b) 81, 9
(c) 256, 16 (d) 625, 25

25. Let a, b, c be in an A.P. Consider the following statements:
I. $\frac{1}{ab},\frac{1}{ca},\frac{1}{bc}$ are in an A.P.
II. $\frac{1}{\sqrt{b}+\sqrt{c}},\frac{1}{\sqrt{c}+\sqrt{a}},\frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P.
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

26. What is the sum of all natural numbers between 200 and 400 which are divisible by 7?
(a) 6729 (b) 8712
(c) 8729 (d) 9276

27. Which term of the sequence 20, $19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}$,….. is the first negative term?
(a) 27^{th} (b) 28^{th}
(c) 29^{th} (d) No such term exists

ANSWER KEY																			
1.	c	**2.**	c	**3.**	b	**4.**	b	**5.**	d	**6.**	c	**7.**	c	**8.**	b	**9.**	b	**10.**	a
11.	a	**12.**	b	**13.**	c	**14.**	b	**15.**	a	**16.**	a	**17.**	c	**18.**	c	**19.**	a	**20.**	c
21.	b	**22**	a	**23.**	d	**24.**	a	**25.**	c	**26.**	c	**27.**	b						

Solutions

Sol.1. (c)
The GP is a, ar, ar^2, ... ar^{2n}
So, P = a. ar. ar^2. ar^3 .... ar^{2n}
$=a^{2n+1}.\ R^{1+2+....+2n}$
$=a^{2n+1}.\ r^{\frac{2n(2n+1)}{2}}$
$=a^{2n+1}\ r^{n(2n+1)} = (ar^n)^{(2n+1)}$
=(2n + 1)th power of the (n + 1) th term of G.P

Sol.2. (c)
If (x + 1), (4x + 1) and (8x + 1) are in GP.
Then $(4x + 1)^2 = (x + 1)(8x + 1)$
$\Rightarrow 16x^2 + 8x + 1 = 8x^2 + x + 8x + 1$
$\Rightarrow 8x^2 - x = 0 \Rightarrow x(8x - 1) = 0$
$\Rightarrow$x =0, 1/8, [1/8 is non-trivial value]

Sol.3. (b)
The given equation
$(a^2 + b^2)x^2 - 2b(a + c)x + (b^2 + c^2) = 0$
Has equal roots, so, discriminate = 0
Hence, $\{2b(a + c)\}^2 - 4(a^2 + b^2)(b^2 + c^2) = 0$
$\Rightarrow 4b^2(a^2 + c^2 + 2ca) - 4(a^2b^2 + a^2c^2 + b^4 + b^2c^2) = 0$
$\Rightarrow b^2a^2 + b^2c^2 + 2b^2ca - a^2b^2 + a^2c^2 - b^4 - b^2c^2 = 0$
$\Rightarrow 2b^2ca = b^4 + a^2c^2$
$\Rightarrow b^4 - 2b^2ca + a^2c^2 = 0$
$\Rightarrow (b^2)^2 - 2(b^2)(ac) + (ac)^2 = 0$
$\Rightarrow (b^2 = ac)^2 = 0$
$\Rightarrow b^2 = ac$
$\Rightarrow$a, b, c are in GP

Sol.4. (b)
Since, first term and common difference of an AP are u and v respectively.
p^{th} term, Tp = u + (p – 1) v .... (i)
and q^{th} term, T_q = u + (q – 1)v .... (ii)
According to condition given in question
$\Rightarrow T_P - T_q + 15uv$
$\Rightarrow T_P - T_q = 15uv$
$\Rightarrow u + (p - 1)v - u - (q - 1)v = 15uv$
$\Rightarrow v(p - 1 - q + 1) = 15uv$
$\Rightarrow v(p - q) = 15uv$
$\Rightarrow p - q = 15u \Rightarrow p = q + 15u$

Sol.5. (d)
Since, a, b, c are in AP
$\frac{1}{a},\frac{1}{b},\frac{1}{c}$
Since, AM ≥ HM
$\Rightarrow b \geq \frac{2ac}{a+c}$
[since AM = b and HM = $\frac{2ac}{a+c}$]
$\Rightarrow ab + bc \geq 2ac$

Sol.6. (c)
As given 1, x, y, z 16are in geometric progression.
Let common ratio be r,
x = 1. r = r
y = 1 . $r^2 = r^2$
z = 1. $r^3 = r^3$
and 16 = 1. $r^4 \Rightarrow 16 = r^4$
$\Rightarrow r = 2$
$\therefore$ x = 1. r = 2, y = 1. r^2 = 4,
z = 1. r^3 = 8
$\therefore$ x + y + z = 2 + 4 + 8 = 14

Sol.7. (c)
As given n_{th} tem
$T_n = 3n + 7$
Sum of n term, $S_n = \Sigma T_n$
$= \Sigma(3n + 7) = 3\Sigma n + 7\Sigma 1$
$= \frac{3n(n+1)}{2} + 7n = n\left[\frac{3n+3+14}{2}\right] = n\left[\frac{3n+17}{2}\right]$
Sum of 50 terms $S_{50} = 50\left[\frac{3\times 50+17}{2}\right]$
$= 50\left[\frac{167}{2}\right] = 25 \times 167 = 4175$

Sol.8. (b)
The given expression $9^{1/3}, 9^{1/9}\ 9^{1/27}$ ... ∞
Can be written as:
$9^{\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+....\infty} = 9^{\frac{\frac{1}{3}}{1-\frac{1}{3}}} = 9^{\frac{1}{3}/\frac{2}{3}} = 9^{1/2} = 3$

Sol.9. (b)

Let first installment be Rs. x and difference of consecutive installments be Rs. d.

$\Rightarrow \frac{30}{2}[2x+29d] = \frac{3600\times 2}{3}$

($\because$ 1/3rd amount is unpaid, 2/3rd amount is paid)

$\Rightarrow 2x+29 = \frac{2400}{15}$

$\Rightarrow 2x + 29d = 160$ ....(1)

Since total amount was 3600 and it was to be paid in 40 installment.

$\Rightarrow \frac{40}{2}$ [2x + 39d]=3600

$\Rightarrow 2x + 39x = 180$ ….(2)

On Solving eqs. (1) and (2), we et

x = 51 and d = 2

First installment = Rs. 51

Sol.10. (a)

If the sequence is in AP with first term, a and common difference, d.

$\Rightarrow T_n = a + (n - 1)d$

Also $T_{n-1} = a + (n - 2)d$

So, the sequence is in AP for which difference between the nth term and (n – 1) th term is independent of n.

Sol.11. (a)

Given $a_n = 2n - 1$

$\therefore S_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n}(2n-1)$

$= 2\sum_{k=1}^{n} n - n = 2.\frac{n(n+1)}{2} - n = n^2 + n - n = n^2$

Sol.12. (b)

Geometric mean = $\sqrt[n+1]{1.2.4.8...2^n}$

($\because$ there are (n+1) multiples from 2° to 2^n)

$= \sqrt[n+1]{2^0.2^1.2^2.2^3......2^n}$

$= \sqrt[n+1]{2^{1+2+3+....+n}} = \sqrt[n+1]{2^{\frac{n(n+1)}{2}}}$

$= \left[2^{\frac{n(n+1)}{2}}\right]^{\frac{1}{n+1}} = 2^{\frac{n}{2}}$

Sol.13. (c)

Let the AP is

a, a + d, a + 2d, ..., a + (2n – 1) d, a + 2nd

Series of even terms.

a + d, a + 3d, ..., a + (2n –1) d, has n terms

Sum of even number= $\frac{n}{2}$

[(a+d)+{a+(2n–1)d}]

$= \frac{n}{2}$[2a+2nd]=n[a+nd]

Series of odd terms

a, a + 2d, a + 4d, ..., a + 2nd has (n + 1 terms.

Sum of odd numbers $= \frac{n+1}{2}[a+(a+2nd)]$

$= \frac{n+1}{2}$ [2a+2d], so, the required ratio= = $\frac{n+1}{n}$

=(n+1(a+nd)

Sol.14. (b)

Given,

$S_n = n^2 - 2n$

$\therefore a_n = S_n - S_{n-1}$

$= n^2 - 2n - [(n-1)^2 - 2(n-1)]$

$= n^2 - 2n - [n^2 + 1 - 2n + 2] = 2n - 3$

Sol.15. (a)

Since, a, 2a + 2 and 3a + 3 are in GP

$\therefore (2a+2)^2 = a(3a+3)$

$\Rightarrow 4a^2+4+8a=3a^2+3a \Rightarrow a^2+5a+4 = 0$

$\Rightarrow a(a+4)+1(a+4)=0 \Rightarrow (a+4)(a+1) = 0$

$\Rightarrow a + 4 = 0$ or $-1 = 0$

$\Rightarrow a = -4$ or -1

Let the fourth term be x.

$\therefore \frac{a}{2a+2} = \frac{3a+3}{x}$

$\Rightarrow x = \frac{(3a+3)(2a+2)}{2}$

When a = – 4

x = –13.5

and a = –1, x = 0

So, the fourth term is –13.5

Sol.16. (a)

Let S = 9 + 99 + 999 + ...

$=(10^1-1)+(10^2-1)+(10^3-1)+.....$

$=(10+10^2+10^3+...)-(1+1+1+....100\text{times})$

$= \frac{10(10^{100}-1)}{10-1} - 100$

($\because 10, 10^2, 10^3$… is GP with a=10, r=10

$S_{100} = \frac{a(r^{100}-1)}{r-1}$ =10/9 $(10^{100}-1)$–100

Sol.17. (c)

11th group is (56, 57, 58, 59, ……66)

Here, a = 56, d = 58 – 57 = 57 – 56 = 1, I=66=T_n

Now, T_n = a (n–1)d

$\Rightarrow 66 = 56 + (n-1)\times 1$

$\Rightarrow n-1=66-56 \Rightarrow n-1 = 10$

$\therefore$n=11

Now, S_n =n/2 [2a + (n–1)d]

S_n =11/2 [2×56+(11–1)×1]

=11/2× [112+10]=11/2×122=11×61=671

Sol.18. (c)

$a + (m-1)d = \frac{1}{n}$

$a + (n-1)d = \frac{1}{m}$

On subtracting, we get

$(m-n)d = \left(\frac{1}{n} - \frac{1}{m}\right) \Rightarrow d = \frac{1}{mn}$

Putting d= $\frac{1}{mn}$ in (i)

We get a = $\frac{1}{mn}$

$\therefore T_{mn} = a + (mn-1)d = \frac{1}{mn} + (mn-1).\frac{1}{mn} = 1$

Sol.19. (a)

Since, $2y^2 = x^2 + z^2$

Let 2(z + x) = (y + z) + (x + y)

$\Rightarrow 2z + 2x = 2y + z + y$

$\Rightarrow z + x = 2y$

If y + z, z + x, x + y are in HP

$\therefore z + x = \frac{2(y+z)(x+y)}{y+z+x+y}$

$\Rightarrow z + x = \frac{2(y+z)(x+y)}{2y+z+x}$

$\Rightarrow 2yz + z^2 + xz + 2xy + xz + x^2$

$= 2yx + 2y^2 + 2zx + 2yz$

$\Rightarrow z^2 + x^2 = 2y^2$

Hence, x^2, y^2 and z^2 are in AP.

Sol.20. (c)

Let the AP is

a, a + d, a + 2d,…., a + (2n–1)d, a +2nd

Series of even terms.

a + d, a + 3d,…., a + (2n–1)d has n term

$\therefore$ Sum =n/2 [a+d]+(a+(2n–1)d)]

=n/2 [2a+2nd] = n[a + nd]

Series of odd terms

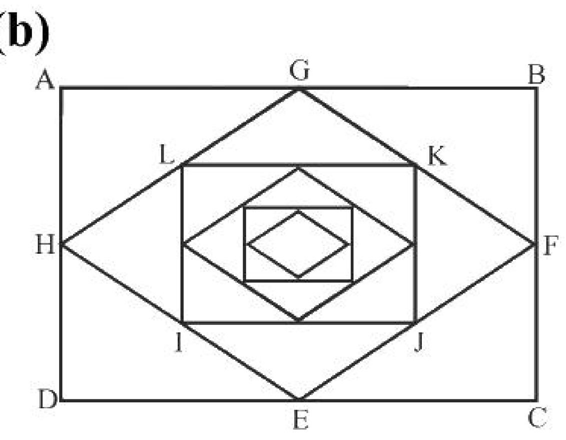
a, a + 2d, a + 4d, …..,a + 2nd has (n+1)terms

$\therefore$ Sum = $\frac{n+1}{2}$ [a+(a+2nd)]

$= \frac{n+1}{2}$ (2a+2nd)=(n+1)(a+nd)

So, the ratio = $\frac{n+1}{2}$

Sol.21. (b)



Required sum

$=16^2 + \frac{1}{2}(16)^2 + \frac{1}{4}(16)^2 + ...\infty$

$=(16)^2 \left(1+\frac{1}{2}+\frac{1}{4}+\infty\right) = 16^2\left(\frac{1}{1-\frac{1}{2}}\right)$

=256×2 = 512sq cm.

Sol.22. (a)

Given, A.G. = m:n

$\Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{m}{n} \Rightarrow \frac{(a+b)^2}{4ab} = \frac{m^2}{n^2}$ …(i)

and $\frac{(a+b)^2 - 4ab}{4ab} = \frac{m^2-n^2}{n^2}$

(using componendo and dividendo rule)

$\Rightarrow \frac{(a-b)^2}{4ab} = \frac{m^2-n^2}{n^2}$

…(ii)

From Eqs. (i) and (ii), we get $\frac{(a+b)^2}{(a-b)^2} = \frac{m^2}{m^2-n^2}$

$\Rightarrow \frac{(a+b)}{(a-b)} = \frac{m}{\sqrt{m^2-n^2}}$

$\Rightarrow \frac{(a+b)+(a-b)}{(a+b)-(a-b)} = \frac{m+\sqrt{m^2-n^2}}{m-\sqrt{m^2-n^2}}$

(using component and dividend rule)

$\Rightarrow \frac{a}{b} = \frac{m+\sqrt{m^2-n^2}}{m-\sqrt{m^2-n^2}}$

Sol.23. (d)

Let a G.P. series is a, ar, ar^2

Where, a = First term and r = common ratio

Sum of infinite series (S) = $\frac{a}{1-r}$

or 6= $\frac{a}{(1-r)}$ or a = 6(1–r)

Sum of first two terms (S_2) = a + ar or 9/2 =a(1+r)

by eq. (i), put the value of a

9/2=(1–r) (1+r) or 3/4 =1–r^2 or r = ±1/2

CaseI. If r = ½, then a = 6 $\left(1-\frac{1}{2}\right)$ or a = 3

Case. II. If r = –2 ½, then $a = 6\left(1+\frac{1}{2}\right)$ or $a = 3 \times 3 = 9$

Sol.24. (a)
Let H be the harmonic mean of two numbers.
$\therefore$ G=H+1.6 and A=H+1.6+2=H+3.6
We know that, $AH = G^2$
$(H+3.6)H = (H+1.6)^2$
$\Rightarrow H^2 + 3.6H = H^2 + 2.56 + 3.2H \Rightarrow H = \frac{256}{0.4} = 6.4$
$\therefore$ A = 6.4 + 3.6=10 and G = 6.4 + 1.6 = 8
Let two numbers are a and b
$\therefore$ a + b = 20
And ab = 64
We know that,
$(a-b)^2 = (a+b)^2 - 4ab = 400 - 256 = 144$
$\Rightarrow a - b = 12$
On solving equations (i) and (iii),we get
a = 16 and b = 4

Sol.25. (c)
Let $\frac{1}{ab}, \frac{1}{ab}, \frac{1}{bc}$ are in AP
$\Rightarrow \frac{1}{ca} - \frac{1}{ab} = \frac{1}{bc} - \frac{1}{ca} \Rightarrow \frac{b-c}{abc} = \frac{a-b}{abc}$
$\Rightarrow$b–c=a–b $\Rightarrow$ 2b = a + c
So, a, b, c are in AP
Now, $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in AP.
$\therefore \frac{2}{\sqrt{c}+\sqrt{a}} = \frac{1}{\sqrt{b}+\sqrt{c}} + \frac{1}{\sqrt{a}+\sqrt{b}}$
$\Rightarrow 2\left(\sqrt{b}+\sqrt{c}\right)\left(\sqrt{a}+\sqrt{b}\right)$
$= \left(\sqrt{c}+\sqrt{a}\right)\left(\sqrt{a}+2\sqrt{b}+\sqrt{c}\right)$
$\Rightarrow 2\left(\sqrt{ab}+b+\sqrt{ac}+\sqrt{bc}\right)$
$= \sqrt{ac}+2\sqrt{bc}+c+a+2\sqrt{ab}+\sqrt{ac}$
$\Rightarrow 2\sqrt{ab}+2b+2\sqrt{ac}+2\sqrt{bc}$
$= 2\sqrt{ac}+2\sqrt{bc}+2\sqrt{ab}+c+a$
$\Rightarrow$2b = a + c
So, a, b, c are in AP
Hence, both the statements are correct.

Sol.26. (c)
The number between 200 and 400 which are divisible by 7 are 203, 210, ….. 399.
$T_n = a + (n-1)d \Rightarrow 399 - 203 + (n-1)7$
$399 = 203 + 7n - 7 \Rightarrow 399 - 203 = 7n - 7$
$196 = 7n - 7 \Rightarrow 203 = 7n \Rightarrow n = 29$
$S_n = n/2\,[29 + (n-1)d] = 29/2[2\times203+(29-1)\times7]$
$=29/2\,[406+28\times7] = 29/2[406+196] =$
$\frac{29}{2}\times602 = 8729$

Sol.27. (b)
Given series can be written as 20, $\frac{77}{4}, \frac{37}{2}, \frac{71}{4}, ...$
This is an AP. Series. Here, a = 20 and d = –3/4
$\therefore T_n = a + (n-1)\,d = 20 + (n-1)\left(-\frac{3}{4}\right) = \frac{83}{4} - \frac{3}{4}n$
For first negative term, $T_n < 0$
$\Rightarrow \frac{83}{4} - \frac{3}{4}n < 0 \Rightarrow 83 < 3n \Rightarrow n > 83/3$
So, n should be 28.
Hence, 28th term is first negative term

NDA PYQ

1. $\frac{1}{b-a}+\frac{1}{b-c}=\frac{1}{a}+\frac{1}{c}$ then a, b, c are in:
(a) A.P. (b) G.P.
(c) H.P. (d) None of these
[NDA - 2011]

2. What is the sum of $\sqrt{3}+\frac{1}{\sqrt{3}}+\frac{1}{3\sqrt{3}}+........?$
(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{3\sqrt{3}}{2}$
(c) $\frac{2\sqrt{3}}{3}$ (d) $\sqrt{3}$
[NDA - 2011(1)]

3. In a GP of positive terms, any term is equal to one-third of the sum of next two terms. What is the common ratio of the GP?
(a) $\frac{\sqrt{13}+1}{2}$ (b) $\frac{\sqrt{13}-1}{2}$
(c) $\frac{\sqrt{13}+1}{3}$ (d) $\sqrt{13}$
[NDA - 2011(1)]

4. Which term of the series $\frac{1}{4},-\frac{1}{2},1,..........$ is –128?
(a) 9^{th} (b) 10^{th}
(c) 11^{th} (d) 12^{th}
[NDA-2011(1)]

5. If n!, 3 × (n!) and (n+1) ! are in G.P. then the value of n will be:
(a) 3 (b) 4
(c) 8 (d) 10
[NDA-2011(2)]

6. What is the 10^{th} common term between the series 2 + 6 + 10 + ….. and 1 + 6 + 11 + ……?
(a) 180 (b) 186
(c) 196 (d) 206
[NDA – 2011(2)]

7. If a, b, c, d, e, f are in AP, then (e–c) is equal to which one of the following?
(a) 2(c–a) (b) 2(d–c)
(c) 2(f–d) (d) (d–c)
[NDA – 2011(2)]

8. If the10th term of a GP is 9 and 4^{th} term is 4, then what is its 7^{th} term?
(a) 6 (b) 14
(c) 27/14 (d) 56/15
[NDA – 2011(2)]

9. What is the nth term of the sequence 1, 5, 9, 13, 17, …..?
(a) 2n – 1 (b) 2n + 1
(c) 4n – 3 (d) None of these
[NDA – 2012(1)]

10. What does the series $1+\frac{1}{\sqrt{3}}+3+\frac{1}{3\sqrt{3}}+.....$ represent?
(a) AP (b) GP
(c) HP (d) None of these
[NDA - 2012(1)]

11. What is the sum of the series $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+.......?$
(a) 1/2 (b) 3/2
(c) 2 (d) 2/3
[NDA - 2012(1)]

12. If 1/4, 1/x and 1/10 are in HP, and then what is the value of x?
(a) 5 (b) 6
(c) 7 (d) 8
[NDA - 2012(1)]

13. If the sequence $\{S_n\}$ is a geometric progression and $S_2S_{11} = S_pS_8$, then what is the value of p?
(a) 1
(b) 3
(c) 5
(d) Cannot be determined
[NDA - 2012(1)]

14. If p, q and r are in AP as well as GP, then which one of the following is correct?
(a) $p = q \neq r$ (b) $p \neq q \neq r$
(c) $p \neq q = r$ (d) $p = q = r$
[NDA - 2012(1)]

15. What is the sum of first eight terms of the series $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+.....?$
(a) $\frac{89}{128}$ (b) $\frac{57}{384}$
(c) $\frac{85}{128}$ (d) None of these
[NDA – 2012(2)]

16. The angles of a triangle are in AP and the least angle is 30°. What is the greatest angle (in radian)?
(a) π/2 (b) π/3
(c) π/4 (d) π
[NDA – 2012(2)]

Direction (for next two)
The sum of first 10 terms and 20 terms of an AP are 120 and 440, respectively.

17. What is its first term?
(a) 2 (b) 3
(c) 4 (d) 1
[NDA – 2012(2)]

18. What is the common difference?
(a) 1 (b) 2
(c) 4 (d) 4
[NDA – 2012(2)]

19. Consider the following statements:
1.The sum of cubes of first 20 natural numbers is 444000
2.The sum of square of first 20 natural numbers is 2870
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2
[NDA-2012(2)]

20. If the numbers n – 3, 4n – 2, 5n + 1 are in AP, what is the value of n?
(a) 1 (b) 2
(c) 3 (d) 4
[NDA – 2013(1)]

21. What is the seventh term of the sequence 0, 3, 8, 15, 24?
(a) 63 (b) 48
(c) 35 (d) 33
[NDA – 2013(2)]

22. The sum of the first five terms and the sum of the first ten terms of an AP are same. Which one of the following is the correct statement?
(a)The first terms must be negative
(b)The common difference must be negative
(c)Either the first term or the common difference is negative but not both
(d)Both the first term and the common difference are negative
[NDA – 2013(2)]

23. What is 0.9 + 0.09 + 0.009 +…. Equal to?
(a) 1 (b) 1.01
(c) 1.001 (d) 1.1
[NDA – 2013(2)]

24. If the positive integers a, b, c and d are in AP, then the numbers abc, abd, acd and bcd are in
(a) HP (b) AP
(c)GP (d) None of these
[NDA – 2013(2)]

25. The sum of an infinite GP is x and the common ratio r is such that $|r| < 1$. If the first term of the GP is 2, then which one of the following is correct?
(a) $-1 < x < 1$ (b) $-\infty < x < 1$
(c) $1 < x < \infty$ (d) None of these
[NDA(I) - 2014]

26. The sum of the series formed by the sequence 3, $\sqrt{3}$, 1, …. up to infinity is
(a) $\frac{3\sqrt{3}\,(\sqrt{3}+1)}{2}$ (b) $\frac{3\sqrt{3}\,(\sqrt{3}-1)}{2}$
(c) $\frac{3(\sqrt{3}+1)}{2}$ (d) $\frac{3(\sqrt{3}-1)}{2}$
[NDA(I) - 2014]

Directions (for next two) Let S_n denotes the sum of first n terms of an AP and $3S_n = S_{2n}$.

27. What is $S_{3n} : S_n$ equal to?
(a) 4 : 1 (b) 6 : 1
(c) 8 : 1 (d) 10 : 1
[NDA(II) - 2014]

28. What is $S_{3n} : S_{2n}$ equal to?
(a) 2 : 1 (b) 3 : 1
(c) 4 : 1 (d) 5 : 1
[NDA(II) - 2014]

29. What is the sum of the series 0.5 + 0.55 + 0.555 + …..+ n terms?
(a) $\frac{5}{9}\left[n-\frac{2}{9}\left(1-\frac{1}{10^n}\right)\right]$ (b) $\frac{1}{9}\left[5-\frac{2}{9}\left(1-\frac{1}{10^n}\right)\right]$
(c) $\frac{1}{9}\left[n-\frac{5}{9}\left(1-\frac{1}{10^n}\right)\right]$ (d) $\frac{5}{9}\left[n-\frac{1}{9}\left(1-\frac{1}{10^n}\right)\right]$
[NDA – 2015(1)]

30. The value of the infinite product $6^{\frac{1}{2}} \times 6^{\frac{1}{2}} \times 6^{\frac{3}{8}} \times 6^{\frac{1}{4}} \times ......$ is
(a) 6 (b) 36
(c) 216 (d) ∞
[NDA – 2015(2)]

31. If the nth term of an AP is $\frac{3+n}{4}$, then the sum of first 105 terms is
(a) 270 (b) 735
(c) 1409 (d) 1470
[NDA – 2015(2)]

32. What is the sum of n terms of the series $\sqrt{2}+\sqrt{8}+\sqrt{18}+\sqrt{32}+....$?
(a) $\frac{n(n-1)}{\sqrt{2}}$ (b) $\sqrt{2}n(n+1)$
(c) $\frac{n(n+1)}{\sqrt{2}}$ (d) $\frac{n(n-1)}{2}$
[NDA – 2015(2)]

33. If p, q, r are in one geometric progression and a, b, c are in another geometric progression, then ap, bq, cr are in
(a)AP (b) GP
(c) HP (d) None of these
[NDA – 2015(2)]

Directions (for next two)
Given that $\log_x y$, $\log_z x$, $\log_y z$ are in GP, xyz = 64 and x^3, y^3, z^3 are in AP.

34. Which one of the following is correct?
x, y and z are
(a) In AP only
(b) In GP only
(c) In both AP and GP
(d) Neither in AP nor in GP
[NDA – 2016(1)]

35. Which one of the following is correct? xy, yz and zx are
(a) In AP only
(b) In GP only
(c) In both AP and GP
(d)Neither in AP nor in GP
[NDA – 2016(1)]

36. How many geometric progressions is/are possible containing 27, 8 and 12 as three of its/ their terms?
(a) One (b) Two
(c) Four (d) Infinitely many
[NDA - 2016(2)]

Directions (for next three):
Consider the following for the next three items that follow:
Let a, x, y, z, b be in AP, where x + y + z = 15. Let a, p, q, r, b be in HP, where $p^{-1}+q^{-1}+r^{-1}=\frac{5}{3}$.

37. What is the value of ab?
(a) 10 (b) 9
(c) 8 (d) 6
[NDA - 2016(2)]

38. What is the value of xyz?
(a) 120 (b) 105
(c) 90 (d) Cannot be determined
[NDA - 2016(2)]

39. What is the value of pqr?
(a) $\frac{35}{243}$ (b) $\frac{81}{35}$
(c) $\frac{243}{35}$ (d) Cannot be determined
[NDA - 2016(2)]

Directions (for next two)
Consider the following next two items that follow:
The sixth terms of an AP is 2 and its common difference is greater than 1.

40. What is the common difference of the AP so that the product of the first, fourth and fifth terms is greatest?
(a) 8/5 (b) 9/5

(c) 2 (d) 11/5

[NDA - 2016(2)]

41. What is the first term of the AP so that the product of the first, fourth and fifth terms is greatest?
(a) –4 (b) –6
(c) –8 (d) –10

[NDA - 2016(2)]

Directions (for next two)
Consider the following for the next two items that follow:
The interior angles of a polygon of n sides are in AP. The smallest angle is 120° and the common difference is 5°.

42. How many possible values can n have?
(a) One (b) Two
(c) Three (d) Infinitely many

[NDA - 2016(2)]

43. What is the largest interior angle of the polygon?
(a) Only 160° (b) Only 195°
(c) Either 160° or 195° (d) Neither 160° nor 195°

[NDA - 2016(2)]

44. What is the greatest value of the positive integer n satisfying the condition
$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+.......+\frac{1}{2^{n-1}}<2-\frac{1}{1000}$?
(a) 8 (b) 9
(c) 10 (d) 11

[NDA – 2016(2)]

45. The fifth term of an AP of n terms, whose sum is $n^2 – 2n$, is:
(a) 5 (b) 7
(c) 8 (d) 15

[NDA (I) - 2017]

46. The sum of all the two-digit odd numbers is:
(a) 2475 (b) 2530
(c) 4905 (d) 5049

[NDA (I) - 2017]

47. The sum of the roots of the equation $x^2 + bx + c = 0$ (where b and c are non–zero) is equal to the sum of the reciprocals of their squares. Then $\frac{1}{c}, b, \frac{c}{b}$ are in:
(a) AP (b) GP
(c) HP (d) None of these

[NDA (I) - 2017]

48. The sum of the roots of the equation $ax^2 + x + c = 0$ (where a and c are non-zero) is equal to the sum of the reciprocals of their squares. Then a, ca^2, c^2 are in:
(a) AP (b) GP
(c) HP (d) None of these

[NDA (I) - 2017]

49. The sum of the first n terms of the series $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+......$ is equal to:
(a) $2^n – n – 1$ (b) $1 – 2^{–n}$
(c) $2^{–n} + n – 1$ (d) $2^n – 1$

[NDA (I) - 2017]

50. What is the sum of the series 0.3 + 0.33 + 0.333 + … n terms?
(a) $\frac{1}{3}\left[n-\frac{1}{9}\left(1-\frac{1}{10^n}\right)\right]$ (b) $\frac{1}{3}\left[n-\frac{2}{9}\left(1-\frac{1}{10^n}\right)\right]$
(c) $\frac{1}{3}\left[n-\frac{1}{3}\left(1-\frac{1}{10^n}\right)\right]$ (d) $\frac{1}{3}\left[n-\frac{1}{9}\left(1+\frac{1}{10^n}\right)\right]$

[NDA (I) - 2017]

51. If the sum of m terms of an AP is n and the sum of n terms is m, then the sum of (m + n) terms is:
(a) mn (b) m + n
(c) 2(m + n) (d) –(m + n)

[NDA (I) - 2017]

52. The value of the product: $6^{\frac{1}{2}}\times6^{\frac{1}{4}}\times6^{\frac{1}{8}}\times6^{\frac{1}{16}}\times....$ up to infinite terms is:
(a) 6 (b) 36
(c) 216 (d) 512

[NDA (II) - 2017]

53. If $S_n = nP+\frac{n(n-1)Q}{2}$, where S_n denotes the sum of the first terms of an AP, then the common difference is:
(a) P + Q (b) 2P + 3Q
(c) 2Q (d) Q

[NDA (II) - 2017]

54. If y = x + x^2 + x^3 + ….. up to infinite terms, where x < 1, then which one of the following is correct?
(a) $x=\frac{y}{1+y}$ (b) $x=\frac{y}{1-y}$
(c) $x=\frac{1+y}{y}$ (d) $x=\frac{1-y}{y}$

[NDA (II) - 2017]

55. A person is to count 4500 notes. Let a_n denote the number of notes the counts in the nth minute. If $a_1 = a_2 = a_3 = …. = a_{10}$ = 150, and a_{10}, a_{11}, a_{12}, …. are in AP with the common difference –2, then the time taken by him to count all the notes is:
(a) 24 min (b) 34 min
(c) 125 min (d) 135 min

[NDA (II) - 2017]

56. If $1.3 + 2.3^2 + 3.3^3 + … + n.3^n = \frac{(2n-1)3^a+b}{4}$, then a and b are respectively:
(a) n, 2 (b) n, 3
(c) n + 1, 2 (d) n + 1, 3

[NDA (II) - 2017]

57. The third term of a GP is 3. What is the product of the first 5 terms?
(a) 216
(b) 226
(c) 243
(d) Can not determined

[NDA (I) - 2018]

58. What is the sum of all two digit numbers which when divided by 3 leaves 2 as the remainder?
(a) 1565 (b) 1585
(c) 1635 (d) 1655

[NDA (I) - 2018]

59. If x, 3/2, z are in AP, x, 3, z are in GP, than which of the following will be in HP.
(a) x, 6, z (b) x, 4, z
(c) x, 2, z (d) x, 1, z

[NDA (I) - 2018]

60. If x = 1 – y + y^2 – y^3 + ….up to infinite terms. Where |y| <1, then which one of the following is correct?
(a) $x=\frac{1}{1+y}$ (b) $x=\frac{1}{1-y}$
(c) $x=\frac{y}{1+y}$ (d) $x=\frac{y}{1-y}$

[NDA (I) - 2018]

61. If a, b, c are in AP or GP or HP, then $\frac{a-b}{b-c}$ is equal to:
(a) $\frac{b}{a}$ or 1 or $\frac{b}{c}$ (b) $\frac{c}{a}$ or $\frac{c}{b}$ or 1
(c) 1 or $\frac{a}{b}$ or $\frac{a}{c}$ (d) 1 or $\frac{a}{b}$ or $\frac{c}{a}$
[NDA (II) - 2018]

62. If the second term of GP is 2 and the sum of its infinite terms is 8, then the GP is:
(a) $8, 2, \frac{1}{2}, \frac{1}{8}$..... (b) $10, 2, \frac{2}{5}, \frac{2}{25}$
(c) $4, 2, 1\frac{1}{2}. \frac{1}{2^2}$,....... (d) $6, 3, \frac{3}{2}, \frac{3}{4}$
[NDA (II) - 2018]

63. Let T_r be the r^{th} term of an AP for r = 1, 2, 3,…If for some distinct positive integers m and n we have $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$ then what is T_{mn} equal to?
(a) $(mn)^{-1}$ (b) $m^{-1} + n^{-1}$
(c) 1 (d) 0
[NDA (II) - 2018]

64. The sum of the series $3 - 1 + \frac{1}{3} - \frac{1}{9} + ...$ is equal to:
(a) 20/9 (b) 9/20
(c) 9/4 (d) 4/9
[NDA (II) - 2018]

65. If an infinite GP has the first term x and the sum 5, then which one of the following is correct?
(a) $x < -10$ (b) $-10 < x < 10$
(c) $0 < x < 10$ (d) $x > 10$
[NDA (II) - 2018]

66. What is the fourth term of an AP of n terms whose sum is n(n+1)?
(a) 6 (b) 8
(c) 12 (d) 20
[NDA-2019(1)]

67. What is n^{th} term of the sequence 25, -125, 625, -3125,....?
(a) $(-5)^{2n-1}$ (b) $(-1)^{2n} 5^{n+1}$
(c) $(-1)^{2n-1} 5^{n+1}$ (d) $(-1)^{n-1} 5^{n+1}$
[NDA (I) - 2019]

68. The number 1, 5 and 25 can be three terms (not necessarily consecutive) of
(a) Only one AP
(b) More than one but finite terms of APs
(c) Infinite number of APs
(d) Finite number of GPs
[NDA (I) - 2019]

69. The sum of $(p+q)^{th}$ and $(p - q)^{th}$ terms of an AP is equal to
(a) $(2p)^{th}$ term (b) $(2q)^{th}$ term
(c) Twice the p^{th} term (d) Twice the q^{th} term
[NDA (I) - 2019]

70. Let a, b, c be in AP and $k \neq 0$ be a real number. Which of following are correct?
1. ka, kb, kc are in AP.
2. k − a , k − b , k − c
3. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}$ are in AP
Select the correct answer using the given code below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
[NDA (II) - 2019]

71. How many two digit numbers are divisible by 4?
(a) 21 (b) 22
(c) 24 (d) 25
[NDA (II) - 2019]

72. Let S_n be the sum of the first n terms of an AP. If $S_{2n} = 3n + 14n^2$, then what is the common difference?
(a) 5 (b) 6
(c) 7 (d) 9
[NDA (II) - 2019]

73. If 3rd, 8th and 13th terms of a GP are p, q, r respectively, then which one of the following is correct?
(a) $q^2 = pr$ (b) $r^2 = pq$
(c) $pqr = 1$ (d) $2q = p + r$
[NDA (II) - 2019]

74. What is the value of $1 - 2 + 3 - 4 + 5 -$ ....+ 101?
(a) 51 (b) 55
(c) 110 (d) 111
[NDA (II) - 2019]

75. If the sum of first n terms of a series is (n + 12), then what is its third term?
(a) 1 (b) 2
(c) 3 (d) 4
[NDA (II) - 2019]

76. A geometric progression (GP) consists of 200 terms. If the sum of odd terms of the GP is m, and the sum of even terms of the GP is n, then what is its common ratio?
(a) m / n (b) n / m
(c) m + (n / m) (d) n + (m / n)
[NDA (II) - 2019]

77. Let m and n (m < n) be the roots of the equation $x^2 - 16x + 39 = 0$, If four terms p, q, r and s are inserted between m and n to form an AP, then what is the value of p + q + r + s ?
(a) 29 (b) 30
(c) 32 (d) 35
[NDA (II) - 2019]

78. If p^2, q^2 and r^2 (where p, q, r > 0) are in G.P. then which of the following is/are correct?
1. p, q and r in GP
2. ln p , ln q, ln r in A.P.
Select the correct answer using the code given below
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA 2020]

79. The third term of a GP is 3. What is the product of first five terms?
(a) 81
(b) 243
(c) 729
(d) Cannot determined
[NDA (I) - 2021]

80. If x^2, x, −8 are in AP then which one of the following is correct?
(a) $x \in \{-2\}$ (b) $x \in \{4\}$
(c) $x \in \{-2, 4\}$ (d) $x \in \{-4, 2\}$
[NDA (I) - 2021]

81. Consider the following statements:
1.If each term of a GP is multiplied by same non-zero number, then the resulting sequence is also a GP.
2.If each term of a GP is divided by same non-zero number, then the resulting sequence is also a GP.
Which one of the following is/are correct?
(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2021]

82. If the first term of an AP is 2 and the sum of the first five terms is equal to the one-forth of the sum of next five terms, then what is the sum of first ten terms

(a) –500 (b) –250
(c) 500 (d) 250

[NDA (I) - 2021]

83. What is C(n,1) + C(n,2)+ C(n,3)+ ....+ C(n,n) equal to ?

(a) $2 + 2^2 + 2^3 + ...+ 2^n$
(b) $1 + 2 + 2^2 + 2^3 + ...+ 2^n$
(c) $1 + 2 + 2^2 + 2^3 + ...+ 2^{n-1}$
(d) $2 + 2^2 + 2^3 + ...+ 2^{n-1}$

[NDA (I) - 2021]

84. Let S_K denote the sum of first k terms of an AP. What is $\frac{S_{30}}{S_{20} - S_{10}}$ equal to?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA (II) 2021]

85. If x, y, z are in GP, then which of the following is/are correct?

1. In (3x), In(3y), In(3z) are in AP
2.xyz + In(x), xyz +In(y), xyz+In(z) are in HP

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) 2021]

86. If p = (1111…up to n digits), then what is the value of $9p^2 + p$?

(a) 10^n p (b) $2p.10^n$
(c) 10^n p–1 (d) 10^n p + 1

[NDA (II) 2021]

87. If $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in HP, then which of the following is/are correct?

1. a, b, c are in AP
2. $(b+c)^2$, $(c+a)^2$ $(a+b)^2$ are in GP

Select the correct answer using the code given below:

(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) 2021]

88. If the sum of the first 9 terms of an AP is equal to sum of the first 11 terms, then what is the sum of the first 20 terms?

(a) 20 (b) 10
(c) 2 (d) 0

[NDA (I) 2022]

89. If the 5^{th} term of an AP is $\frac{1}{10}$ and its 10^{th} term is $\frac{1}{5}$, then what is the sum of first 50 terms?

(a) 25 (b) 25.5
(c) 26 (d) 26.5

[NDA (I) 2022]

90. If a, b, c are in GP where a > 0, b > 0, c > 0, then which of the following are correct?

1. a^2, b^2, c^2 are in GP
2. $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in GP
3. $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in GP

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (I) 2022]

91. If $\frac{a+b}{2}, b, \frac{b+c}{2}$ are in HP, then which one of the following is correct?

(a) a, b, c are in AP
(b) a, b, c are in GP
(c) a + b, b + c, c + a are in GP
(d) a + b, b + c, c + a are in AP

[NDA (I) 2022]

Consider the following for the next three (03) items that follow:

Let f (x) be a function satisfying f (x + y) = f (x) f (y) for all x, y ∈ N such that f (1) = 2.

92. If $\sum_{x=2}^{n} f(x) = 2044$, then what is the value of n?

(a) 8 (b) 9
(c) 10 (d) 11

[NDA 2022 (II)]

93. What is $\sum_{x=1}^{5} f(2x-1)$ equal to?

(a) 341 (b) 682
(c) 1023 (d) 1364

[NDA 2022 (II)]

94. What is $\sum_{x=1}^{6} 2^x f(x)$ equal to

(a) 1365 (b) 2730
(c) 4024 (d) 5460

[NDA 2022 (II)]

95. Consider the following statements:

1.2 + 4 + 6 +….. + 2n = n^2 + n
2.The expression $n^2 + n + 41$ always given a prime number for every natural number n.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:

Let P be the sum of first n positive terms of an increasing arithmetic progression A, Let Q be the sum of first n positive terms of another increasing arithmetic progression B.
Let P : Q = (5n + 4):(9n + 6)

96. What is the ratio of the first term of A to that of B?

(a) 1/3 (b) 2/5
(c) 3/4 (d) 3/5

[NDA 2022 (II)]

97. What is the ratio of their 10^{th} terms?

(a)11/29 (b) 22/49
(c) 33/59 (d) 44/69

[NDA 2022 (II)]

98. If d is the common difference of A, and D is the common difference of B, then which one of the following is always correct?

(a) D > d
(b) D < d
(c) 7D > 12d
(d) None of the above

[NDA 2022 (II)]

99. p, q, r and s are in AP such that p + s = 8 and qr = 15. What is the difference between largest and smallest numbers?

(a) 6 (b) 5
(c) 4 (d) 3

[NDA – 2023 (1)]

100. If $2^{\frac{1}{c}}, 2^{\frac{b}{ac}}, 2^{\frac{1}{a}}$ are in GP, then which one of the following is correct?
(a) a, b, c are in AP (b) a, b, c are in GP
(c) a, b, c are in HP (d) ab, bc, ca are in AP
[NDA – 2023 (1)]

101. The first and the second terms of an AP are $\frac{5}{2}$ and $\frac{23}{12}$ respectively. If nth term is the largest negative term, what is the value of n?
(a) 5
(b) 6
(c) 7
(d) n cannot be determined
[NDA – 2023 (1)]

102. In an AP, the first term is x and the sum of the first n terms is zero. What is the sum of next m terms?
(a) $\frac{mx(m+n)}{n-1}$ (b) $\frac{mx(m+n)}{1-n}$
(c) $\frac{nx(m+n)}{m-1}$ (d) $\frac{nx(m+n)}{1-m}$
[NDA – 2023 (1)]

103. If the n^{th} term of a sequence is $\frac{2n+5}{7}$, then what is the sum of its first 140 terms?
(a) 2840 (b) 2780
(c) 2920 (d) 5700
[NDA-2023 (2)]

104. If (a+b), 2b, (b+c) are in HP, then which one of the following is correct?
(a) a, b and c are in AP
(b) a–b, b–c and c–a are in AP
(c) a, b and c are in GP
(d) a–b, b–c and c–a are in GP
[NDA-2023 (2)]

105. Let t_1, t_2, t_3 ….. be in GP. What is $(t_1t_3.....t_{21})^{\frac{1}{11}}$ equal to?
(a) t_{10} (b) t_{10}^2
(c) t_{11} (d) t_{11}^2
[NDA-2023 (2)]

Consider the following for the next (02) items that follow
Let a_1, a_2, a_3…… be in AP such that
$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{25} + a_{30} + a_{34} = 300$.

106. What is $a_1 + a_5 - a_{10} - a_{15} - a_{20} - a_{25} + a_{30} + a_{34}$ equal to?
(a) 0 (b) 25
(c) 125 (d) 250
[NDA-2023 (2)]

107. What is $\sum_{n=1}^{34} a_n$ equal to:
(a) 900 (b) 1025
(c) 1200 (d) 1275
[NDA-2023 (2)]

108. If a, b, c are in HP, then what is $\frac{1}{b-a}+\frac{1}{b-c}$ equal to ?
1. $\frac{2}{b}$
2. $\frac{1}{a}+\frac{1}{c}$
c. $\frac{1}{2}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Select the correct answer using the code given below
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3
[NDA-2024 (1)]

109. If a, b, c are in AP; b, c, d are in GP; c, d, e are in HP, then which of the following is/are correct?
1. a, c and e are in GP
2. $\frac{1}{a},\frac{1}{c},\frac{1}{e}$ in GP
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below
The product of 5 consecutive terms of an AP is 229635. The first, second and fifth terms are in GP.

110. What is the common difference?
(a) 3 (b) 4
(c) 5 (d) 6
[NDA-2024 (2)]

111. What is the sum of all five terms?
(a) 60 (b) 65
(c) 75 (d) 80
[NDA-2024 (2)]

112. In an AP, the ratio of the sum of the first p terms to the sum of the first q terms is $p^2 : q^2$. Which one of the following is correct ?
(a) The first term is equal to the common difference
(b) The first term is equal to twice the common difference
(c) The common difference is equal to twice the first term
(d) The first term is equal to square of the common difference
[NDA-2024 (2)]

113. Let x > 1, y > 1, z > 1 be in GP.
Then $\frac{1}{1+\ln x},\frac{1}{1+\ln y},\frac{1}{1+\ln z}$ are
(a) in AP
(b) in GP
(c) in HP
(d) neither in AP nor in GP nor in HP
[NDA-2024 (2)]

114. If the sum of the first n terms of a series is n(2n + 1), then what is nth term?
(a) 4n – 1 (b) 4n
(c) 4n + 1 (d) 4n + 3
[NDA-2024 (2)]

115. If p times the pth term of an AP is equal to q times of qth term (p ≠ q), then what is the (p + q)th term equal to ?
(a) 0 (b) p + q
(c) pq (d) pq(p + q)
[NDA-2024 (2)]

116. Let $p = \ln(x)$, $q = \ln(x)^3$ and $r = \ln(x)^5$, where x > 1. Which of the following statements is/are correct?
I. p, q and r in AP
II. p, q and r can never be in GP
Select the correct answer using the code given below ?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
[NDA-2024 (2)]

117. The sum of the first *k* terms of a series *S* is $3k^2 + 5k$. Which one of the following is correct?
(a) The terms of *S* form an arithmetic progression with common difference 14.

(b) The terms of S form an arithmetic progression with common difference 6.
(c) The terms of S form a geometric progression with common ratio 10/7.
(d) The terms of S form a geometric progression with common ratio 11/4.
[NDA-2025 (1)]

118. The sum of the first 8 terms of a GP is five times the sum of its first 4 terms. If $r \neq 1$ is the common ratio, then what is the number of possible real values of r?
(a) One (b) two
(c) Three (d) More than three
[NDA-2025 (1)]

119. If 5th, 7th and 13th terms of an AP are in GP, then what is the ratio of its first term to its common difference?
(a) –3 (b) –2
(c) 2 (d) 3
[NDA-2025 (1)]

120. If p, 1, q are in AP and p, 2, q are in GP, then which of the following statements is/are correct?
I. p, 4, q are in HP.
II. $(1/p)$, 1/4, $(1/q)$ are in AP.
Select the answer using the code given below:
(a) I only (b) II only
(c) Both I and II (d) neither I nor II
[NDA-2025 (1)]

121. If $p^x = q^y = r^z$, where x, y and z are in GP, then consider the following statements:
I. p, q and r are in AP.
II. $\ln p$, $\ln q$ and $\ln r$ are in GP.
Which of the statements given above is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
[NDA-2025 (2)]

122. If $(10 + \log_{10} x)$, $(10 + \log_{10} y)$ and $(10 + \log_{10} z)$ are in AP, then consider the following statements:
I. The GM of x and z is y^2.
II. The AM of $\log_{10} x$ and $\log_{10} z$ is $\log_{10} y$.
Which of the statements given above is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
[NDA-2025 (2)]

123. How many terms of the series 1 + 3 + 5 + 7 + ... amount to a sum equal to 12345678987654321?
(a) 11111111 (b) 11000011
(c) 11110111 (d) 111111111
[NDA-2025 (2)]

124. How many terms are identical in the two APs 19, 21, 23, ... up to 110 terms and 19, 22, 25, 28, ... up to 75 terms?
(a) 35 (b) 36
(c) 37 (d) 38
[NDA-2025 (2)]

For the following two (02) items:
Let (6 + 10 + 14 + ……… up to m terms) = (1 + 3 + 5 + 7 + …….. up to n terms), where (m < 25) and (n < 25.)

125. What is the relation between m and n?
(a) $n^2 = m(m + 1)$ (b) $n^2 = m(m + 2)$
(c) $n^2 = 2m(m + 1)$ (d) $n^2 = 2m(m + 2)$
[NDA-2025 (2)]

126. How many values of m are possible?
(a) None (b) One
(c) Two (d) More than two
[NDA-2025 (2)]

ANSWER KEY

1.	c	**2.**	b	**3.**	b	**4.**	b	**5.**	c	**6.**	b	**7.**	b	**8.**	a	**9.**	c	**10.**	d
11.	d	**12.**	c	**13.**	c	**14.**	d	**15.**	c	**16.**	a	**17.**	b	**18.**	b	**19.**	b	**20.**	a
21.	b	**22**	c	**23.**	a	**24.**	a	**25.**	c	**26.**	a	**27.**	b	**28.**	a	**29.**	d	**30.**	b
31.	d	**32.**	c	**33.**	b	**34.**	c	**35.**	c	**36.**	d	**37.**	b	**38.**	b	**39.**	c	**40.**	a
41.	b	**42.**	a	**43.**	a	**44.**	c	**45.**	b	**46.**	a	**47.**	c	**48.**	a	**49.**	c	**50.**	a
51.	d	**52.**	a	**53.**	d	**54.**	a	**55.**	b	**56.**	d	**57.**	c	**58.**	c	**59.**	a	**60.**	a
61.	c	**62.**	c	**63.**	c	**64.**	c	**65.**	c	**66.**	b	**67.**	d	**68.**	c	**69.**	c	**70.**	d
71.	b	**72.**	c	**73.**	a	**74.**	a	**75.**	a	**76.**	b	**77.**	c	**78.**	c	**79.**	b	**80.**	c
81.	c	**82.**	b	**83.**	c	**84.**	c	**85.**	a	**86.**	a	**87.**	a	**88.**	d	**89.**	b	**90.**	d
91.	b	**92.**	c	**93.**	b	**94.**	d	**95.**	c	**96.**	d	**97.**	c	**98.**	a	**99.**	a	**100.**	a
101.	b	**102.**	b	**103.**	c	**104.**	c	**105.**	c	**106.**	a	**107.**	d	**108.**	*	**109.**	c	**110.**	d
111.	c	**112.**	c	**113.**	c	**114.**	a	**115.**	a	**116.**	c	**117.**	b	**118.**	c	**119.**	a	**120.**	c
121.	b	**122.**	b	**123.**	d	**124.**	c	**125.**	d	**126.**	c								

Solutions

Sol.1. (c)

$$\frac{1}{b-a}+\frac{1}{b-c}=\frac{1}{a}+\frac{1}{c}$$

$$\frac{1}{b-a}-\frac{1}{c}=\frac{1}{a}-\frac{1}{b-c}$$

$$\frac{c-b+a}{c(b-a)}=\frac{b-c-a}{a(b-c)}$$

$$\frac{1}{c(b-a)}=\frac{-1}{a(b-c)}$$

$$2ac = bc + ab$$

$$\frac{2}{b}=\frac{1}{a}+\frac{1}{c}$$

a, b, c are in H.P.

Sol.2. (b)

$\sqrt{3}+\frac{1}{\sqrt{3}}+\frac{1}{3\sqrt{3}}+........$

$S_\infty=\frac{a}{1-r}=\frac{\sqrt{3}}{1-\frac{1}{3}}=\frac{3\sqrt{3}}{2}$

Sol.3. (b)
Let three terms of GP are a, ar, ar^2

$a=\frac{1}{3}\left(ar+ar^2\right)$

$3a=\left(ar+ar^2\right)$

$ar+ar^2=3a$

$r+r^2=3$

$r^2+r-3=0$

$r=\frac{-1\pm\sqrt{1+12}}{2}=\frac{-1+\sqrt{13}}{2}$

r > 0 so we will take only 1 value of r.

Sol.4. (b)

$\frac{1}{4}-\frac{1}{2}+1............-128$

$T_n=ar^{n-1}$

$-128=\frac{1}{4}(-2)^{n-1}$

$-512=(-2)^{n-1}$

n – 1 = 9
n = 10

Sol.5. (c)
Given, that
n!, 3 × (n!) and (n+1)! are in GP, then using
$b^2 = ac$
$\{3 \times (n!)\}^2 = (n!) \times (n + 1)!$
$\Rightarrow (3)^2 (n!) = (n+1)!$
{as we know that n! = n × (n–1)!}
⇒9(n!) = (n+1). (n!)
⇒9= n+1
∴n=8

Sol.6. (b)
Let the series;
S_1=2+6+10+14+18+22+26+30+34+38+42+46+
S_2=1+6+11+16+21+26+31+36+41+46+…..
The common number of the series S_1 and S_2 is
S = 6 + 26 + 46 + …..
∴T_{10} = 6 + (10 – 1) × 20 = 186

Sol.7. (b)
Given, a, b, c, d, e and f are in AP., then
Difference(D) = b – a = c – b = d – c = e – d
e – c = twice of difference = 2(d-c)

Sol.8. (a)
Given, 10th term of GP = 9 …(i)
$T_{10} = ar^{(10-1)} \Rightarrow ar^9 = 9$
and $T_4 = ar^{(4-1)} = ar^3 = 4$ …(ii)
multiply both equation
$a^2r^{12} = 36$
$ar^6 = 6 = 7^{th}$ term

Sol.9. (c)
$T_n = a + (n-1)d$ (n^{th} term)
=1 + (n–1) 4 = 1 + 4n – 4 = 4n –3

Sol.10. (d)
between any two consecutive terms, no common difference or common ratio is formed.
Hence, the given series does not form any progression.

Sol.11. (d)
∴Sum of infinite term of GP =

$\frac{a}{1-r}=\frac{1}{1-\left(\frac{-1}{2}\right)}=\frac{1}{1+1/2}=\frac{2}{3}$

Sol.12. (c)
Given that, ¼, 1/x and 1/10 are in HP
⇒4, x and 10 are in AP, then
2x = 4+10
x = 7

Sol.13. (c)
We know that, in a GP the product of two terms equidistant from the beginning and end is a constant and is equal to the product of first term and last term, i.e. if:
$a_1, a_2, a_3, …..a_{(n-2)}, a_{(n-1)}$, an are in GP, then
$a_1a_n = a_2\ a_{n-1} = a_3\ a_{n-2} = …..$
Given that,
$S_2S_{11} = S_pS_8 \Rightarrow (p + 8)= (2 + 11)$
p = 13 – 8 = 5

Sol.14. (d)
If 3 numbers are in AP as well as in GP
This is only possible when all these numbers are equal.

Sol.15. (c)
∴The sum of first eight terms of the series

$=\frac{a\left(1-r^8\right)}{1-r}=\frac{1\left[1-\left(-\frac{1}{2}\right)^8\right]}{1-\left(-\frac{1}{2}\right)}=\frac{1-\frac{1}{256}}{1+\frac{3}{2}}=\frac{25}{128}$

Sol.16. (a)
30°+(30°+ d)+(30°+2d)=180°
3d = 90°
d = 30°
Greatest angle = 90° or π/2

Sol.17. (b)
Given, S_{10} = 120 and S_{20} =440
∵S_n =n/2 [2a + (n–1)d]
⇒120=5(2a+9d)
⇒ 2a + 9d = 24 ……….(i)
⇒440 = 10 (2a + 19d)
⇒2a + 19d = 44 ……..…(ii)
On subtracting Eq. (i) from Eq. (ii), we get
10d = 20 ⇒ d = 2
On putting the value of d in Eq. (i), we get
2a + 9(2) = 24
⇒2a + 18 = 24
⇒2a = 6
⇒ a = 3

Sol.18. (b)
By above solution d = 2

Sol.19. (b)
Sum of cube of numbers

$=\left[\frac{n(n+1)}{2}\right]^2=\left[\frac{20(21)}{2}\right]^2=44100$

Sum of square of numbers

$=\left[\frac{n(n+1)(2n+1)}{6}\right]=\left[\frac{20(21)(41)}{6}\right]=2870$

Only statement 2 is correct.

Sol.20. (a)
Given that, (n–3), (4n–2), (5n+1) are in AP
∴(4n–2) – (n–3) = (5n+1) – (4n–2)
⇒3n+1 = n+3
⇒2n=2 ⇒ n=1

Sol.21. (b)
Difference of series is increasing by 2
So series is 0, 3, 8, 15, 24, 35, 48………
Seventh term is 48

Sol.22. (c)
Let a be the first term and d be the common difference of an AP.
$S_5 = S_{10}$

$\Rightarrow \frac{5}{2}(2a+4d)=\frac{10}{2}(2a+9d)$

⇒ a+2d =2a + 9d
⇒ a+7d = 0
∴a = – 7d …(i)
Let first five terms of an AP are a – 2d, a –d, a, a + d, and a + 2d.
⇒ – 9d, –8d, –7d, –6d, –5d
Here, first term = – 9d
and common difference = – 8d + 9d = d
So, if d is positive, then first term should be negative and common difference should be positive.
If d is negative, then first term should be positive and common difference should be negative.
Hence, either the first term or the common difference is negative, but not both.

Sol.23. (a)
Given series, =0.9 + 0.09 + 0.009 + …..
[infinite GP with common ratio = 1/10]

$\frac{9}{10}\times\frac{1}{\left(1-\frac{1}{10}\right)}=\frac{9}{10}\times\frac{10}{9}=1$

Sol.24. (a)
Given that, a, b, c and d are in AP
$\Rightarrow \frac{1}{a},\frac{1}{b},\frac{1}{c},\frac{1}{d}$ are in HP
Multiply by abcd
⇒bcd, acd, abd and abc are in HP
Hence, abc, abd, acd and bcd are in HP.

Sol.25. (c)
Given that,
The sum of an infinite GP = x

$x=\frac{2}{1-r}$

$r=1-\frac{2}{x}$

For infinite GP

$-1<r<1$

$-1<1-\frac{2}{x}<1$

$-2<-\frac{2}{x}<0$

$1<x<\infty$

Sol.26. (a)
∴Sum of infinite GP, =

$\frac{a}{1-r}=\frac{2}{1-\frac{1}{\sqrt{3}}}$

$\frac{3\sqrt{3}\left(\sqrt{3}+1\right)}{3-1}=\frac{3\sqrt{3}\left(\sqrt{3}+1\right)}{2}$

Sol.27. (b)
We, have
Sn = Sum of first n terms of an AP

$S_n=\frac{n}{2}[2a+(n-1)d]$

Similarly,

$S_{2n}=\frac{2n}{2}[2a+(2n-1)d]$

Now, $3S_n = S_{2n}$

$\frac{3n}{2}[2a+(n-1)d]=\frac{2n}{2}[2a+(2n-1)d]$

⇒ 2a = d(n + 1)
so

$\therefore S_n=\frac{n}{2}[dn+1(+d(n-1)]=n^2 d$

$S_{3n}=\frac{3n}{2}[(n+1+3n-1)]=6n^2 d$

$\therefore \frac{S_{3n}}{S_n} = \frac{6n^2d}{n^2d} = 6:1$

Sol.28. (a)

$\frac{S_{3n}}{S_{2n}} = \frac{6n^2d}{3n^2d} = 2:1$

Sol.29. (d)

0.5 + 0.55 + 0.555 +…..upto n terms

$= \frac{5}{10} + \frac{55}{100} + \frac{555}{1000} + ...\text{upto n terms}$

$= \frac{5}{10}\left[1 + \frac{11}{10} + \frac{111}{100} + ...\text{n terms}\right]$

$= \frac{5}{10} \times \frac{1}{9}\left[9 + \frac{99}{10} + \frac{999}{100} + ...\text{upto n terms}\right]$

$= \frac{5}{90}\left[(10-1) + \frac{(10^2-1)}{10} + \frac{(10^3-1)}{10^2} + ...\text{n terms}\right]$

$= \frac{5}{90}\left[10 + \frac{10^2}{10} + \frac{10^3}{10^2} + ...\text{upto n terms}\right]$

$+ \frac{5}{90}\left[-1 - \frac{1}{10} - \frac{1}{10^{-2}} - \frac{1}{10^{-3}} - ......\text{upto n terms}\right]$

$= \frac{5}{90}[10 + 10 + 10 + ...\text{upto n terms}]$

$+ \frac{5}{90}(-1)\left[1 + \frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + ...\text{upto n terms}\right]$

$= \frac{5}{90} \times 10n - \frac{5}{90}\left[\frac{1 - \left(\frac{1}{10}\right)^n}{1 - \frac{1}{10}}\right]$

$= \frac{5n}{9} - \frac{5}{90} \times \frac{10}{9}\left[\frac{10^n - 1}{10^n}\right]$

$= \frac{5}{9}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$

Sol.30. (b)

$6^{\frac{1}{2}} \times 6^{\frac{1}{2}} \times 6^{\frac{3}{8}} \times 6^{\frac{1}{4}} \times .....\infty$

$= 6^{\frac{1}{2}} \times 6^{\frac{2}{4}} \times 6^{\frac{3}{8}} \times 6^{\frac{4}{16}} \times .....\infty$

$= 6^{\left(\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + ...\infty\right)}$

$= 6^2 = 36$

Sol.31. (d)

Then nth term of AP. Is $\frac{3+n}{4}$

Put n = 1 then $T_1 = 1$

Put n = 2 then $T_2 = 5/4$

So difference = ¼

$S_{105} = \frac{105}{2}\left[2 + 104\left(\frac{1}{4}\right)\right] = 1470$

Sol.32. (c)

$\sqrt{2} + \sqrt{8} + \sqrt{18} + \sqrt{32} + ...$ upto n terms

$= \sqrt{2}$ [1 + 2 + 3 + 4 + …. upto n terms]

$= \frac{\sqrt{2}\left[n(n+1)\right]}{2} = \left\{\frac{n(n+1)}{\sqrt{2}}\right\}$

Sol.33. (b)

p, q, r are in GP. $\Rightarrow q^2 = pr$ …(i)

a, b, c are in GP. $\Rightarrow b^2 = ac$ …(ii)

Now, $b^2 q^2 = apcr$ {from (i) & (ii)}

Or $(bq)^2 = (ap).(cr)$

$\therefore$ ap, bq, cr are in GP.

Sol.34. (c)

$\log_x y, \log_z x, \log_y z$ are in GP

$(\log_z x)^2 = \log_x y.\log_y z$

$\left(\frac{\log x}{\log z}\right)^2 = \frac{\log y}{\log x}.\frac{\log z}{\log y}$

$\left(\frac{\log x}{\log z}\right)^2 = \frac{\log z}{\log x}$

$\left(\frac{\log x}{\log z}\right)^3 = 1$

x = z …………(i)

x^3 , y^3 , z^3 are given in AP

$2y^3 = x^3 + z^3$

$2y^3 = 2x^3$

x = y …………(ii)

so x = y = z

so x,y,z are in AP and GP

Sol.35. (c)

if x = y = z then xy = yz = zx

so these are also in both AP and GP

Sol.36. (d)

Let a be the first term and r be the common ratio.

Let 27, 8, 12 be the pth, qth, tth terms of the GP.

$\therefore$ 27 = ar^{P-1}

8 = ar^{q-1}

and 12 = ar^{t-1}

We know, 27 × 8^2 = 12^3

$\therefore$ ar^{p-1} $(ar^{q-1})2$ = $(ar^{t-1})3$

$\Rightarrow$ r^{p-1} $(r^{q-1})2$ = $(rt^{-1})3$

$\Rightarrow$ p – 1 + 2q – 2 = 3t – 3

$\Rightarrow$ p + 2q – 3t = 0

There are infinite solutions for the equation p + 2q – 3t = 0.

$\therefore$ There are infinite number of geometric progression possible.

Sol.37. (b)

Given, a, x, y, z, b are in AP.

$2y = x + z = a + b$ …….(i)

given that x + y + z = 15

so 3y = 15

y = 5

by eq (i)

a + b = 10 ………(ii)

a, p, q, r, b be in HP, where $p^{-1}+q^{-1}+r^{-1}=\frac{5}{3}$.

$\frac{1}{a}, \frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{b}$ are in AP

so

$\frac{2}{q} = \frac{1}{r} + \frac{1}{p} = \frac{1}{a} + \frac{1}{b}$ ………..(iii)

given that

$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{5}{3}$

by eq (iii)

$\frac{3}{q} = \frac{5}{3}$

$q = \frac{9}{5}$

so

$= \frac{1}{a} + \frac{1}{b} = \frac{10}{9} = \frac{a+b}{ab}$ [$\because$ a+b=10]

$\Rightarrow$ ab = 9

Sol.38. (b)

We have,

1, x, y, z, 9 are in AP.

i.e. x = 3 , y = 5 , z = 7

$\therefore$ xyz = 7 × 5 × 3 = 105

Sol.39. (c)

Since, 1, p, q, r, 9 are in HP

$1, \frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{9}$ are in AP

$\frac{9}{9}, \frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{9}$ are in AP

$\frac{9}{9}, \frac{7}{9}, \frac{5}{9}, \frac{3}{9}, \frac{1}{9}$ are in AP

so $pqr = \frac{9.9.9}{7.5.3} = \frac{243}{35}$

Sol.40. (a)

Given, a + 5d = 2 and d > 1

$\therefore$ Product of first, fourth and fifth terms

P(d) = a × (a + 3d) × (a + 4d)

=(2 – 5d)(2 – 5d + 3d)(2 – 5d + 4d)

=(2 – 5d) (2 – 2d)(2 – d)

= 8 – 32d + 34d^2 – 10d^3

P'(d) = – 32 + 68d – 30d^2

[Differentiating w.r.t.x]

For greatest value of P (d), P'(d) = 0

$\Rightarrow$–32 + 68d – 30d2 = 0 $\Rightarrow$15d2 – 34d + 16 = 0

$\Rightarrow$(3d – 2) (5d – 8) = 0 $\Rightarrow$d =2/3 and 8/5

Now, P''(d) = 68 – 60d [differentiating again]

for d=2/3

P''(d) = 68 – 60 × 2/3 =28 > 0

for d =8/5,

P''(d) = 68 – 60 ×8/5 = – 28 < 0

Hence, for d =8/5

P''(d)has the greatest value.

Sol.41. (b)

When, d= 8/5 , then product of first, fourth and firth terms is maximum

and we have a + 5d = 2 $\Rightarrow$ a + 5 ×8/5 =2

$\therefore$ a = 2 – 8 = – 6.

Sol.42. (a)

$\frac{n}{2}$ [2a + (n–1)d]=(n–2)×180

$\Rightarrow \frac{n}{2}$ (2 × 120 + (n–1)(5)]=(n – 2) × 180

$\Rightarrow \frac{n}{2}$ [(2 × 24 +(n – 1)]=(n – 2) × 36

$\Rightarrow$n(48 + n –1) = (n – 2) × 72

$\Rightarrow$n(n + 47) = 72n – 144

$\Rightarrow n^2$ + 47n – 72n + 144 = 0

$\Rightarrow n^2$ – 25 + 144 = 0

$\Rightarrow$(n – 16)(n – 9) = 0

$\Rightarrow$ n= 9, 16

Here, only n = 9 is the required number sides.

Since, if n = 16, then angle 195° > 180°, which is not possible.

Sol.43. (a)

Largest angle = a_9 = a + 8d

=120° + 8 × 5° = 120° + 40° = 160°

Sol.44. (c)

We have,

$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + ... + \frac{1}{2^{n-1}} < 2 - \frac{1}{1000}$

$\Rightarrow \frac{1\left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} < 2 - \frac{1}{1000}$

$2\left(1 - \frac{1}{2^n}\right) < 2 - \frac{1}{1000}$

$\left(2 - \frac{2}{2^n}\right) < 2 - \frac{1}{1000}$

$\left(-\frac{2}{2^n}\right) < -\frac{1}{1000}$

$$\left(\frac{2}{2^n}\right) > \frac{1}{1000}$$

$$\left(\frac{1}{2^{n-1}}\right) > \frac{1}{1000}$$

$\Rightarrow 2^{n-1} < 1000$ $\{\because 2^{10} = 1024 > 1000\}$

$\therefore n \le 10$

Sol.45. (b)

$S_n = n^2 - 2n$

$\therefore T_n = S_n - S_{n-1}$

$= (n^2 - 2n) - [(n-1)^2 - 2(n-1)]$

$= n^2 - 2n - (n^2 - 2n + 1 - 2n + 2)$

$= n^2 - 2n - n^2 + 2n - 3 + 2n = 2n - 3$

$\Rightarrow T_5 = 2 \times 5 - 3 = 7.$

Sol.46. (a)

Sn = 11 + 13 + 15 + ……. + 99

$$= \frac{45}{2}[11+99] \qquad \left\{\because \frac{n}{2}(a+l)\right\}$$

$$= \frac{45}{2} \times 110 = 45 \times 55 = 2475.$$

Sol.47. (c)

Let α and β be the roots of the equation x2 + bx + c = 0, then

$$\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\alpha + \beta = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$-b = \frac{(-b)^2 - 2c}{(c)^2}$$

$$-b = \frac{b^2 - 2c}{c^2}$$

$-bc^2 = b^2 - 2c$

$$\frac{2c}{b + c^2} = b \qquad \text{….(i)}$$

Now, if $\frac{1}{c}, b, \frac{c}{b}$ are in H.P., then

$$b = \frac{2 \times \frac{1}{c} \times \frac{c}{b}}{\frac{1}{c} + \frac{c}{b}} = \frac{2c}{c^2 + b} \qquad \text{….(ii)}$$

Clearly from equations (i) and (ii),

$\frac{1}{c}, b, \frac{c}{b}$

are on H.P.

Sol.48. (a)

If α and β are the roots of the equation $ax^2 + x + c = 0$.

$$\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\alpha + \beta = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$-\frac{1}{a} = \frac{\left(-\frac{1}{a}\right)^2 - 2\frac{c}{a}}{\left(\frac{c}{a}\right)^2}$$

$$-\frac{1}{a} = \frac{1 - 2ca}{c^2}$$

$-c^2 = a - 2ca^2$

$2ca^2 = a + c^2$

Clearly, a, ca^2, c^2 are in A.P.

Sol.49. (c)

$$S_n = \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{5}{16} + \text{.....n terms}$$

$$\Rightarrow T_n = \frac{2^n - 1}{2^n} = 1 - \frac{1}{2^n}$$

$$\Rightarrow \sum_{n=1}^{n=n} T_n = \sum_{n=1}^{n=n}(1) - \sum_{n=1}^{n=n}\frac{1}{2^n}$$

$$= n - \left[\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + ... + \frac{1}{2^n}\right]$$

$$= n - \frac{\frac{1}{2}\left[1 - \frac{1}{2^n}\right]}{1 - \frac{1}{2}}$$

$$= n - 1 + \frac{1}{2^n}.$$

Sol.50. (a)

Sn = 0.3 + 0.33 + 0.333 + …. n terms

$$= \frac{3}{10} + \frac{33}{100} + \frac{333}{1000} + \text{....n terms}$$

$$= 3\left[\frac{1}{10} + \frac{11}{10^2} + \frac{111}{10^3} + \text{.....n terms}\right]$$

$$= \frac{3}{9}\left[\frac{9}{10} + \frac{99}{10^2} + \frac{999}{10^3} + \text{....n terms}\right]$$

$$= \frac{1}{3}\left[\frac{10-1}{10} + \frac{100-1}{10^2} + \frac{1000-1}{10^3} + \text{.....n terms}\right]$$

$$= \frac{1}{3}\left[1 - \frac{1}{10} + 1 - \frac{1}{10^2} + 1 - \frac{1}{10^3} + \text{...n terms}\right]$$

$$= \frac{1}{3}\left[(1+1+1+\text{.....n terms}) - \left(\frac{1}{10} + \frac{1}{10^2} + \text{....n terms}\right)\right]$$

$$= \frac{1}{3}\left[n - \left\{\frac{\frac{1}{10}\left(1 - \frac{1}{10^n}\right)}{1 - \frac{1}{10}}\right\}\right]$$

$$= \frac{1}{3}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$$

Sol.51. (d)

Remember:

If $S_m = n$

and $S_n = m$

then $S_{m+n} = -(m+n)$

Sol.52. (a)

$$6^{\frac{1}{2}} \times 6^{\frac{1}{4}} \times 6^{\frac{1}{8}} \times 6^{\frac{1}{16}} \times \text{.......}\infty$$

$$= 6^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\text{......}\infty}$$

$$= 6^{\left(\frac{\frac{1}{2}}{1-\frac{1}{2}}\right)} = 6$$

Sol.53. (d)

$$S_n = nP + \frac{n(n-1)Q}{2}$$

$$= nP + \frac{n^2Q}{2} - \frac{Qn}{2} = \frac{Q}{2}n^2 + \left(P - \frac{Q}{2}\right)n$$

$\therefore$ Common difference=2×coefficient of n^2

$$= 2 \times \frac{Q}{2} = Q$$

Sol.54. (a)

$y = x + x^2 + x^3 + \text{…….}\infty, x < 1.$

$$y = \frac{x}{1-x}$$

{Sum of G.P. of infinite terms}

$\Rightarrow y - xy = x$

$\Rightarrow xy + x = y$

$$x = \frac{y}{1+y}$$

Sol.55. (b)

$a_1 = a_2 = a_3 = \text{…….} = a_{10} = 150$

$\therefore$ He counted 9 × 150 = 1350 notes in 9 minutes.

Now, a_{10}, a_{11}, a_{12} ….., are in AP

Such that $a_{10} = 150$ and difference = –2

$\therefore$ 150 + 148 + 146 +…..= 4500 – 1350 = 3150

$$\frac{n}{2}[150 \times 2 + (n-1) \times (-2)] = 3150$$

or $150n - n^2 + n = 3150$

or $150\,n - n^2 + n = 3150$

or n2 – 151n + 3150 = 0

or (n – 126n) n (n – 25) = 0

or n = 25 or 126

Clearly, n = 25 minutes

$\therefore$ The time taken by him to count all the notes = 25 + 9 = 34 minutes.

Sol.56. (d)

$S = 1 \times 3 + 2 \times 3^2 + 3 \times 3^3 + \text{….} + n \times 3^n$ ……..(i)

or $3S = 1 \times 3^2 + 2 \times 3^3 + \text{…} + (n-1) \times 3^n + n \times 3^{n+1}$ …..(ii)

$(1 - 3)\,S = 1 \times 3 + 1 \times 3^2 + \text{….. …..} + 1 \times 3^n - n \times 3^{n+1}$

$$-2 \times S = \frac{3(3^n - 1)}{3 - 1} - n \times 3^{n+1}$$

$$S = \frac{-3(3^n - 1)}{4} + \frac{n \times 3^{n+1}}{2}$$

$$= \frac{-3^{n+1} + 3 + 2n \times 3^{n+1}}{4}$$

$$= \frac{(2n-1).3^{n+1} + 3}{4}$$

or $$\frac{(2n-1).3^a + b}{4} = \frac{(2n-1).3^{n+1} + b}{4}$$

$\therefore$ On comparing we get, a=n+1 and b=3.

Sol.57. (c)

third term = $ar^2 = 3$

product of first 5 terms = $a.ar.ar^2.ar^3.ar^4$

$= a^5r^{10} = (ar^2)^5 = 3^5 = 243$

Sol.58. (c)

Sum = 11 +14 + ………..….+ 98

$$\frac{(11+98)}{2} \times 30$$

= 1635

Sol.59. (a)

x+z = 3, xz = 9

$$\frac{2xz}{x+z} = \frac{18}{3} = 6$$

$\Rightarrow$x, 6, z $\in$ HP

Sol.60. (a)

$$S_\infty = \frac{a}{1-r}$$

$$x = \frac{1}{1-(-y)} = \frac{1}{1+y}$$

Sol.61. (c)

If a, b, c is in AP

$$b = \frac{a+c}{2}$$

$$\frac{a-b}{b-c} = \frac{a - \frac{a+c}{2}}{\frac{a+c}{2} - c} = \frac{\frac{2a-a-c}{2}}{\frac{a+c-2c}{2}} = \frac{a-c}{a-c} = 1$$

If a, b, c is in GP

$b^2 = ac \Rightarrow c = b^2/c$

$$\frac{a-b}{b-c}=\frac{a-b}{b-\frac{b^2}{a}}=\frac{(a-b)a}{ab-b^2}=\frac{(a-b)}{ab-b^2}=\frac{a}{b}$$

If a, b, c are in HP

$$b=\frac{2ac}{a+c}$$

$$\frac{a-b}{b-c}=\frac{a-\frac{2ac}{a+c}}{\frac{2ac}{a+c}-c}=\frac{\frac{a(a+c)-2ac}{a+c}}{\frac{2ac-c(a+c)}{a+c}}$$

$$\frac{a^2+ac-2ac}{2ac-ac-c^2}=\frac{a^2-ac}{ac-c^2}$$

$$\frac{a(a-c)}{c(a-c)}=\frac{a}{c}$$

Sol.62. (c)

$ar = 2$ ………(i)

$8=\frac{a}{1-r}$ ……(ii)

$8-8r = a$

$(8-8r)r = 2$

$8r - 8r^2 = 2$

$8r^2 - 8r + 2 = 2$

$4r^2 - 4r + 1 = 0$

$(2r-1)^2 = 0$

$r = ½$

so $a = 4$

then series is 4, 2, 1, ½,…….

Sol.63. (c)

$T_m=\frac{1}{n}=a+(m-1)d$ ………(i)

$T_n=\frac{1}{m}=a+(n-1)d$ ………(ii)

by solving above equations

$d=\frac{1}{mn}, a=\frac{1}{mn}$

so $T_{mn}=a+(mn-1)d=\frac{1}{mn}+(mn-1)\frac{1}{mn}=1$

Sol.64. (c)

$$S_\infty=\frac{a}{1-r}=\frac{3}{1-\frac{1}{3}}=\frac{9}{4}$$

Sol.65. (c)

Given that,

The sum of an infinite GP = 5

$5=\frac{x}{1-r}$

$r=1-\frac{x}{5}$

For infinite GP

$-1<r<1$

$-1<1-\frac{x}{5}<1$

$-2<-\frac{x}{5}<0$

$-10 < -x < 0$

$0 < x < 10$

Sol.66. (b)

$S_n = n^2 + n$

$S_1 = 2$

$S_2 = 4 + 2 = 6$

so $a_1 = 2$ and $a_2 = 4$ so $d = 2$

$a_4 = 2 + (4-1)2 = 8$

Sol.67. (d)

25, -125, 625, -3125, ...

this is a GP of ratio -5

$Tn = ar^{n-1}$

$Tn = 25(-5)^{n-1}$

$Tn = (-1)^{n-1}5^{n+1}$

Sol.68. (c)

1,2,3,4,5,6............24,25,26,27

this AP contain 1, 5, 125

difference of this AP is 1.

1,1.5,2,2.5,3,.....,4.5,5,5.5,......24.5,25,25.5,..

this AP also contain 1, 5, 125

difference of this AP is 0.5

so infinite number of AP and GPs are possible.

Sol.69. (c)

$T_{p+q}+T_{p-q}=(a+(p+q-1)d)+(a+(p-q-1)d)$

$=2a + 2pd - 2d$

$=2(a + (p-1)d)$

= twice of pth term

Sol.70. (d)

After addition, subtraction, multiplication, division by any constant AP remains in AP.

Sol.71. (b)

$T_n = a + (n - 1)d$

$96 = 12 + (n - 1) 4$

$n = 22$

Sol.72. (c)

put n = 1

$S_2 = 3 + 14 (1) = 17$

$17=\frac{2}{2}[2a+(2-1)d]$

$2a + d = 17$ ……….(i)

put n = 2

$S_4 = 6 + 14(2)2 = 62$

$62=2[2a+(4-1)d]$

$2a + 3d = 31$ ……..…(ii)

by (ii) − (i)

$2d=14$

$d = 7$

Sol.73. (a)

$Tn = ar^{n-1}$

$p = ar^2$ ....(i)

$q = ar^7$ ....(ii)

$r = ar^{12}$ ....(iii)

by these equations $q^2 = pr$

Sol.74. (a)

(1 − 2) + (3 − 4) + (5 − 6) .... + 101

(− 1) + (− 1) +(− 1)+ ....50 times + 101

= − 50 + 101 = 51

Sol.75. (a)

$S_3 = 15$ and $S_2 = 14$

$T_3 = S_3 - S_2 = 15 - 14 = 1$

Sol.76. (b)

$S_{200} = a + ar + ar^2 + ar^3 + ...ar^{198} + ar^{199}$

$m = a + ar^2 + ar^4 + ar^6 + ... ar^{198}$

$n = ar + ar^3 + ar^5 + ar^7 + ... ar^{199}$

$n = r(a + ar^2 + ar^4 + ar^6 + ... ar^{198}$

$n = rm$

$r = n/m$

Sol.77. (c)

Roots of equation $x^2 - 16x + 39 = 0$

are 3 and 13

m = 3 and n = 13

then p, q, r, s are 5, 7, 9, 11 respectively.

p + q + r + s = 5 + 7 + 9 + 11 = 32

Sol.78. (c)

if three numbers are in GP then their square roots are also in GP

if three numbers are in GP then their logarithm is in AP

Sol.79. (b)

third term $= ar^2 = 3$

product of first 5 terms $= a.ar.ar^2.ar^3.ar^4$

$= a^5r^{10} = (ar^2)^5 = 3^5 = 243$

Sol.80. (c)

x^2 ,x , −8 are in AP (2b = a + c)

$2x = x^2 - 8$

$x^2 - 2x - 8 = 0$

$x = 4, -2$

Sol.81. (c)

both statements are correct.

Sol.82. (b)

$S_5=\frac{1}{4}(S_{10}-S_5)$

$S_{10}=5S_5$

$\frac{10}{2}[2(2)+(10-1)d]=5.\frac{5}{2}[2(2)+(5-1)d]$

$[4+9d]=\frac{5}{2}[4+4d]$

$[8+18d]=5[4+4d]$

$2d = -12$

$d = -6$

$S_{10}=\frac{10}{2}[2(2)+(10-1)(-6)]=-250$

Sol.83. (c)

$C(n,1) + C(n,2)+ C(n,3)+ ....+ C(n,n) = 2^n - 1$

check option 3rd

$1 + 2 + 2^2 + 2^3 + ...+ 2^{n-1} = 2^n - 1$

Sol.84. (c)

$$\frac{S_{30}}{S_{20}-S_{10}}=\frac{\frac{30}{2}[29+29d]}{\frac{20}{2}[29+19d]-\frac{10}{2}[29+9d]}$$

$$=\frac{6a+87d}{4a+38d-29-9d}=\frac{6a+87d}{2a+29d}=3$$

Sol.85. (a)

x, y, z are in GP

$y^2 = xz$

$\log y^2 = \log xz$

$2\log y = \log x + \log z$

Log x, logy, logz are in AP.

xyz + logx, xyz + logy, xyz + log z are in AP

x, y, z are in GP

3x, 3y, 3z are in GP

log 3x, log 3y, log 3z will be in AP.

Sol.86. (a)

P = (1111 …….up to n digits)

$9p^2 + p = p (9p + 1)$

=p[99999 ……up to n digits + 1]

=p[10000…… n zeroes]

$=p(10^n) = 10^n p$

Sol.87. (a)

$\frac{1}{b+c},\frac{1}{c+a},\frac{1}{a+b}$

are in H.P

⇒ b + c, c + a, a + b are in A.P

As we know if a, b, c are in A.P

Then 2b = a + c

⇒2(c + a) = a + b + b + c

2c + 2a = 2b + a + c

c + a = 2b

⇒ a, b, c are in A.P.

So, (1) statement is true

For second part As b + c, c + a, a + b are

In A.P. ⇒ b + c = 1, c + a = 2, a + b = 3

When we solve $(b + c)^2 = 1$, $(c + a)^2 = 4$, $(a + b)^2 = 9$

1,4,9 are not in G.P. so statement is not true.

Sol.88. (d)

$S_9 = S_{11}$

$\frac{9}{2}[2a + 8d] = \frac{11}{2}[2a + 10d]$

$18a + 72d = 22a + 110d$

4a + 38d = 0
2a+ 19d = 0
$S_{20} = \frac{20}{2}$ [2a + 19d] = 0

Sol.89. (b)
$T_n = a + (n-1)d$
$\frac{1}{10} = a + 4d$ …(1)
$\frac{1}{5} = a + 9d$ …(2)
By (1) & (2)
$5d = \frac{1}{10} \Rightarrow d = \frac{1}{50}$
$\Rightarrow a = \frac{1}{50}$
$S_{50} = \frac{50}{2}\left[\frac{2}{50} + \frac{49}{50}\right]$
=25.5

Sol.90. (d)
a, b, c are in GP
then a^2, b^2, c^2 and $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ *and* $\sqrt{a}, \sqrt{b}, \sqrt{c}$
all will be in GP.

Sol.91. (b)
$\frac{a+b}{2}, b, \frac{b+c}{2}$ are in HP
$\frac{2}{b} = \frac{2}{a+b} + \frac{2}{b+c}$
$\frac{1}{b} - \frac{1}{a+b} = \frac{1}{b+c}$
$\frac{a+b-b}{b(a+b)} = \frac{1}{b+c}$
$ab + ac = ab + b^2 =$
$b^2 = ac = ab + b^2$
$\Rightarrow$a, b, c are in GP.

Sol.92. (c)
f(x+y) = f(x) f(y)
given that f(1) = 2
put x= y = 1
$f(2) = [f(1)]^2 = 4$
put x = 2, y = 1
f(3) = f(2)f(1) = 8
put x= y = 2
$f(4) = [f(4)]^2 = 16$
and so on
f(2) + f(3) + f(4) + f(5) + …………. + f(n)
4 + 8 + 16 + 32 + ………… = 2044
[given]
f(1) + f(2) + f(3) + f(4) + f(5) + ………… + f(n) = 2046
2 + 4 + 8 + 16 + 32 + ………… = 2046
$\frac{2(2^n - 1)}{2-1} = 2046$
$2^{n+1} = 2048$
n = 10

Sol.93. (b)
using above solution
f(1) + f(3) + f(5) + f(7) + f(9)
2 + 8 + 32 + 128 + 512 = 682

Sol.94. (d)
using above solution
2f(1) + 4f(2) + 8f(3) + 16f(4) + 32f(5) + 64f(6)
4 + 16 + 64 + 256 + 1024+ 4048 = 5460

Sol.95. (c)
2 + 4 + 6 + 8 + ………. + 2n
=2(1 +2 + 3 + 4 + 5 + 6 ……..... + n)
$2\frac{n(n+1)}{2} = n^2 + n$
statement 1 is correct.
$n^2 + n + 41$ is always prime for any value of n
statement 2 is also correct.

Sol.96. (d)
let first term of AP , A is a_1 and difference is d_1.
let first term of AP , B is a_2 and difference is d_2

$$\frac{P}{Q} = \frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{5n+4}{9n+6}$$
$$= \frac{[2a_1 + (n-1)d_1]}{[2a_2 + (n-1)d_2]} = \frac{5n+4}{9n+6}$$
put n= 1
$$= \frac{a_1}{a_2} = \frac{5+4}{9+6} = \frac{3}{5}$$

Sol.97. (c)
by above solution
$$= \frac{[2a_1 + (n-1)d_1]}{[2a_2 + (n-1)d_2]} = \frac{5n+4}{9n+6}$$
$$= \frac{\left[a_1 + \frac{(n-1)d_1}{2}\right]}{\left[a_2 + \frac{(n-1)d_2}{2}\right]} = \frac{5n+4}{9n+6}$$
……………..(i)
for 10^{th} term $T_{10} = a + 9d$
$\frac{(n-1)}{2} = 9$
n = 19
put this value of n in equation (i)
$$= \frac{[a_1 + 9d_1]}{[a_2 + 9d_2]} = \frac{5(19)+4}{9(19)+6} = \frac{99}{177} = \frac{33}{59}$$

Sol.98. **(a)**
by above solutions we can compare that difference of AP A is less than difference of AP B
so d < D

Sol.99. **(a)**
p, q, r, s are in AP
so p + s = q + r = 8
qr = 15
so q = 3 and r = 5
then p will be 1 and s will be 7
difference between 7 and 1 is 6.

Sol.100. (a)
$2^{\frac{1}{c}}, 2^{\frac{b}{ac}}, 2^{\frac{1}{a}}$ are in GP
so $2^{\frac{2b}{ac}} = 2^{\frac{1}{c}}.2^{\frac{1}{a}}$
$\frac{2b}{ac} = \frac{1}{a} + \frac{1}{c}$
2b = a + c
i.e. a,b,c are in AP

Sol.101. (b)
first term = 5/2 or 30/12
2^{nd} term = 23/12
3^{rd} term = 16/12
4^{th} term = 9/12
5^{th} term = 2/12
6^{th} term = -5/12

Sol.102. (b)
first term = x
$$S_n = 0 = \frac{n}{2}[2x + (n-1)d]$$
$$d = \frac{2x}{1-n}$$
$$S_n = S_{m+n} = \frac{m+n}{2}[2x + (m+n-1)d]$$
$$= S_{m+n} = \frac{m+n}{2}\left[2x + (m+n-1)\frac{2x}{1-n}\right]$$
$$= S_{m+n} = \frac{m+n}{1}\left[\frac{mx}{1-n}\right] = \frac{mx(m+n)}{1-n}$$

Sol.103. **(c)**
$$T_n = \frac{2n+5}{7}$$
put n = 1 then $T_1 = 1$
put n = 2 them $T_2 = 9/7$
difference = 2/7
$$S_{140} = \frac{140}{2}\left[2 + (139)\frac{2}{7}\right] = 2920$$

Sol.104. (c)
a+b, 2b, b+c are in HP
$$\frac{2}{2b} = \frac{1}{a+b} + \frac{1}{b+c}$$
$$\frac{1}{b} = \frac{a+2b+c}{(a+b)(b+c)}$$
$$ab + ac + bc + b^2 = ab + 2b^2 + bc$$
$$b^2 = ac$$
a,b,c are in GP

Sol.105. **(c)**
t_1 , t_2 , t_3 , ………….t_n are in GP
$$(t_1.t_3.t_5.t_7.............t_{21})^{\frac{1}{11}}$$
$$= (a.ar^2ar^4.ar^6...............ar^{20})^{\frac{1}{11}}$$
$$= (a^{11}r^{110})^{\frac{1}{11}} = ar^{10} = t_{11}$$

Sol.106. **(a)**
$a_1 + a_5 - a_{10} - a_{15} - a_{20} - a_{25} + a_{30} + a_{34}$
= a + [a+4d] – [a+9d] – [a+14d] – [a+19d] – [a+24d] + [a+29d] + [a+33d] =0

Sol.107. **(d)**
given that
$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{25} + a_{30} + a_{34}$
a + [a+4d] + [a+9d]+[a+14d] + [a+19d] + [a+24d] + [a+29d] + [a+33d]
= 8a + 132d = 300 ……….(i)
$a_1 + a_2 + a_3$ ……………….. + a_{34}
= a + [a+d] + [a+2d] + [a+3d] + ………+ [a+33d] =34a + 561d = 4.25(8a+132d) = 4.25 x 300 = 1275

Sol.108. **(*)**
a,b,c are in HP
so $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$ or $b = \frac{2ac}{a+c}$
given
$$\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{\frac{2ac}{a+c} - a} + \frac{1}{\frac{2ac}{a+c} - c}$$
$$= \frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$
so statement 1 and 2 are correct.
that is not given in option

Sol.109. **(b)**
a, b, c are in AP
let a = 2, b = 4 , c = 6
b, c, d are in GP
if b = 4 and c = 6 then d = 9
c, d, e are in HP
if c = 6, d = 9 then e = 18
2,6,18 are in GP
and reciprocal of GP is GP

Sol.110. **(d)**
let 5 terms of AP are
a, a + d, a+ 2d, a + 3d, a + 4d
first, second and fifth terms are in GP
$(a + d)^2 = a(a + 4d)$
$a^2 + 2ad + d^2 = a^2 + 4ad$
$d^2 = 2ad$
d = 2a ……….(i)
Given that
(a)(a + d)(a+ 2d)(a + 3d)(a + 4d) = 229635
a.3a.5a.7a.9a = 229635
$a^5 = 243$
a = 3
so common difference = 2a = 6

Sol.111. **(c)**

by above solution
five terms will be 3, 9, 15, 21, 27
sum of above five terms = 75

Sol.112. (c)
according to question

$\frac{S_p}{S_q} = \frac{p^2}{q^2}$

$\Rightarrow \frac{p^2}{q^2} = \frac{\frac{p}{2}[2a+(p-1)d]}{\frac{q}{2}[2a+(q-1)d]}$

$\Rightarrow \frac{p}{q} = \frac{[2a+(p-1)d]}{[2a+(q-1)d]}$

$\Rightarrow 2ap+(q-1)pd = 2aq+(p-1)qd$

$\Rightarrow 2a(p-q)+pqd-d(p-q) = pqd$

$\Rightarrow 2a(p-q)-d(p-q) = 0$

$\Rightarrow (2a-d)(p-q) = 0$

$\Rightarrow 2a = d$

common difference = twice of first term

Sol.113. (c)
x, y, z are in GP
ln x, ln y, ln z will be in AP
1 + ln x, 1 + ln y, 1+ ln z will be in AP
$\frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ will be in HP.

Sol.114. (a)
$S_n = n(2n+1)$
$T_n = S_n - S_{n-1}$
$T_n = n(2n+1) - (n-1)[2(n-1)+1]$
$= 2n^2 + n - (n-1)(2n-1)$
$= 2n^2 + n - 2n^2 + n + 2n - 1$
$= 4n - 1$

Sol.115. (a)
According to question
$pT_p = qT_q$
$p[a + (p-1)d] = q[a + (q-1)d]$
$ap - aq + p^2d - q^2d - pd + qd = 0$
$a(p-q) + d(p^2 - q^2) - d(p-q) = 0$
$(p-q)[a + d(p+q) - d] = 0$
$[a + d(p+q) - d] = 0$
$[a + d(p+q-1)] = 0$
$T_{p+q} = 0$

Sol.116. (c)
$p = \ln(x)$, $q = \ln(x)^3$ and $r = \ln(x)^5$,
we know that
1, 3, 5 are in AP
multiply by ln x
ln x, 3ln x, 5ln x are in AP
statement I is correct.
ln x, 3ln x, 5ln x will be in GP when ln x = 0 that is at x = 1
but according to question x > 1.
so p, q, r can never be in GP
Statement II is also correct.

Sol.117. (b)
$S_k = 3k^2 + 5k$
put k = 1
$S_1 = 3 + 5 = 8$
Put k = 2
$S_2 = 12 + 10 = 22$
Put k = 3
$S_3 = 27 + 15 = 42$
i.e. $T_1 = S_1 = 8$
$T_2 = S_2 - S_1 = 22 - 8 = 14$
$T_3 = S_3 - S_2 = 42 - 22 = 20$
This is an AP of common difference 6.

Sol.118. (c)

$S_n = \frac{a(r^n-1)}{r-1}$

$S_8 = 5S_4$

$\frac{a(r^8-1)}{r-1} = 5\frac{a(r^4-1)}{r-1}$

$(r^8-1) = 5(r^4-1)$

$(r^4-1)(r^4+1) = 5(r^4-1)$

$(r^4-1)(r^4-4) = 0$

$r = 1, -1, \sqrt{2}, -\sqrt{2}$
but $r \neq 1$ number of real values are three.

Sol.119. (a)
$T_5 = a + 4d$
$T_7 = a + 6d$
$T_{13} = a + 12d$
Above three terms are in GP
$(a + 6d)^2 = (a + 4d)(a + 12d)$
$a^2 + 12ad + 36d^2 = a^2 + 16ad + 48d^2$
$4ad + 12d^2 = 0$
$a + 3d = 0$
$a/d = -3$

Sol.120. (c)
p, 1, *q* are in AP
so p + q = 2 ………(i)
p, 2, *q* are in GP
so pq = 4 …….(ii)
if *p*, k, *q* are in HP

$k = \frac{2pq}{p+q} = \frac{2(4)}{2} = 4$

so *p*, 4, *q* are in HP.
and *(1/p)*, 1/4, *(1/q)* are in AP.

Sol.121. (b)
Let $p^x = q^y = r^z = k$
By taking log

$x\log p = \log k \Rightarrow x = \frac{\log k}{\log p}$

$y\log q = \log k \Rightarrow y = \frac{\log k}{\log q}$

$z\log r = \log k \Rightarrow z = \frac{\log k}{\log r}$

x, y and z are in GP
$y^2 = xz$

$\left(\frac{\log k}{\log q}\right)^2 = \left(\frac{\log k}{\log p}\right)\left(\frac{\log k}{\log r}\right)$

$(\log q)^2 = (\log p)(\log r)$

ln p, *ln q* and *ln r* are in GP.
Only statement 2 is correct.

Sol.122. (b)
$10 + \log_{10} x$), $(10 + \log_{10} y)$ and $(10 + \log_{10} z)$ are in AP
$\log_{10} x$, $\log_{10} y$, $\log_{10} z$ are in AP
so AM of $\log_{10} x$ and $\log_{10} z$ is $\log_{10} y$
By taking logarithm of GP series, AP obtained.
i.e. x, y, z are in GP
y is GM of x and z
statement I is wrong, II is correct.

Sol.123. (d)
We know that sum of first n odd numbers is n^2
1 + 3 + 5 +…. Up to n terms = 1234567654321
$n^2 = 1234567654321$
n = 111111111

Sol.124. (c)
110^{th} term of 19, 21, 23, 25, …….
= 19 + (110 – 1)2 = 237
75^{th} term of 19, 22, 25, 28, 31,………
= 19 + (75 – 1)3 = 241
first common term in both series is 19, then 25, 31, and so on with common difference 6, and last common term of this series is 235
235 = 19 + (n – 1)6
n = 37

Sol.125. (d)
(6 + 10 + 14 + ……… up to m terms) = (1 + 3 + 5 + 7 + …….. up to n terms),
$S_m = S_n$

$\frac{m}{2}[12+(m-1)4] = \frac{n}{2}[2+(n-1)2]$

$8m + 4n^2 = 2n^2$
$n^2 = 2m(m+2)$

Sol.126. (c)
By above solution
$n^2 = 2m(m+2)$
put m = 2, then n = 4
put m = 16 then n = 24
two values of m are possible.

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